Bellman Calibration for V-Learning in Offline Reinforcement Learning

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Abstract

We introduce *Iterated Bellman Calibration*, a simple, model-agnostic, post-hoc procedure for calibrating off-policy value predictions in infinite-horizon Markov decision processes. Bellman calibration requires that states with similar predicted long-term returns satisfy the Bellman equation on average under the target policy. We adapt classical histogram and isotonic calibration to the dynamic, counterfactual setting by repeatedly regressing fitted Bellman targets onto a model's predictions, using a doubly robust pseudo-outcome to handle off-policy data. This yields a one-dimensional fitted value iteration scheme that can be applied to any value estimator. Our analysis establishes finite-sample guarantees for Bellman calibration and shows how calibration error translates into value error under weak assumptions, without requiring Bellman completeness or realizability.

1. Introduction

Many applications require predicting the long-term consequences of a decision policy in a sequential, stochastic environment. We study this problem in a Markov decision process (MDP), where the goal is to predict the long-run return under a target policy π, possibly from data generated by a different behavior policy. This off-policy value-prediction problem arises in many domains: clinicians anticipate long-term health outcomes under proposed treatment rules (Van Calster et al., 2019); online platforms estimate customer lifetime value and retention under alternative recommendation strategies (Theocharous et al., 2015; Maystre & Russo, 2022; Xue et al., 2025); and economists assess the downstream effects of counterfactual programs (Rust, 1987; Cowgill & Tucker, 2019). In such settings, practitioners often need more than correct ranking: they need numerically accurate value predictions, since decisions may depend on the magnitude of predicted returns, not just their order. Calibrated value predictions are therefore important for personalized forecasting, long-term prediction, and reliable policy evaluation, since systematic miscalibration can distort policy comparisons and degrade decision-making.

Modern machine learning predictors such as neural networks and gradient-boosted trees are often miscalibrated: their numerical predictions can systematically deviate from realized outcomes due to model misspecification, limited data, or distribution shift (Zadrozny & Elkan, 2001; Niculescu-Mizil & Caruana, 2005; Bella et al., 2010; Guo et al., 2017). This issue is especially important in reinforcement learning, where value estimators may suffer from overestimation, instability, and off-policy bias (Thrun & Schwartz, 2014; Van Hasselt et al., 2016; Fujimoto et al., 2019). A natural requirement is therefore that value predictions be *calibrated*: states assigned similar predicted returns should, on average, realize long-run outcomes consistent with those predictions under the target policy. Calibrated value estimates can improve the reliability and interpretability of personalized forecasting and policy evaluation. While calibration is well studied in classification and regression, where predictions are matched to empirical frequencies or conditional means (Lichtenstein et al., 1977; Platt et al., 1999; Zadrozny & Elkan, 2001; Vovk et al., 2005), it remains comparatively underexplored for long-horizon value prediction in reinforcement learning.

Long-horizon value prediction also poses challenges absent from standard supervised learning. First, it is inherently counterfactual: the goal is to predict returns under a target policy π, often from data generated by a different behavior policy. Second, long-run returns are not directly observed; instead, one must infer them from one-step transitions through a model or the Bellman equation. Third, errors can propagate through the dynamics, so even small modeling inaccuracies may compound over time (Gordon, 1995; Munos & Szepesvári, 2008; Farahmand et al., 2010; Agarwal et al., 2019). Finally, common value-learning procedures such as temporal-difference methods and fitted value or Q-iteration are themselves iterative approximation schemes, so finite-iteration error, function approximation error, and optimization instability can all distort the resulting value estimates (Tsitsiklis & Van Roy, 1996; Fujimoto et al., 2019; Agarwal et al., 2021). As a result, even widely used value estimators may produce systematically biased or unstable long-horizon predictions.

These empirical difficulties reflect deeper theoretical limitations. Classical analyses of fitted value and Q-iteration typically rely on strong assumptions such as Bellman completeness, realizability, or low-dimensional state spaces

(Baird et al., 1995; Tsitsiklis & Van Roy, 1996; Munos & Szepesvári, 2008; Fan et al., 2020; Hu et al., 2025). Bellman completeness ensures that each Bellman target lies within the function class, so every regression step is well specified. When completeness fails, the Bellman target falls outside the class, the regression becomes misspecified, and projection errors can accumulate across iterations—in some cases exponentially with the horizon—leading to arbitrarily poor estimates (Chen & Jiang, 2019; Wang et al., 2021; Foster et al., 2021; Amortila et al., 2020). Recent work relaxes Bellman completeness using min–max or adversarial formulations, but these approaches typically require highly expressive critics, dual realizability, and strong coverage assumptions (Uehara et al., 2021; 2020; Jin et al., 2021; Uehara et al., 2023).

Contributions. We study post-hoc correction of long-horizon value predictors for a target policy π. In large state spaces, full Bellman consistency is often too strong to enforce. We therefore focus on a weaker calibration criterion and a post-hoc procedure for improving value predictions using only the scalar score induced by an initial predictor.

Our main contributions are:

1. We formalize **Bellman calibration**, a minimal correctness criterion for infinite-horizon value prediction that is weaker than Bellman consistency. It is satisfied by the true value function and provides a dynamic analogue of classical calibration.

2. We introduce **post-hoc calibration methods** based on histogram binning and isotonic regression. These can be applied to any learned value predictor, including those produced by fitted value or Q-iteration, and rely on a doubly robust Bellman target for off-policy data.

3. We establish **finite-sample guarantees** for both calibration error and value error. The bounds separate statistical, nuisance-estimation, and finite-iteration errors, and show that our procedure reduces Bellman calibration error without sacrificing the best accuracy achievable within the score class induced by the original predictor. They do not require Bellman completeness or realizability assumptions.

Related work.

Our work extends classical post-hoc calibration ideas from supervised learning to the dynamic, sequential setting (Zadrozny & Elkan, 2001; 2002; Niculescu-Mizil & Caruana, 2005; Bella et al., 2010; Guo et al., 2017); for a modern treatment, see (Roth, 2022). Whereas these works study calibration of static predictions, we study calibration of long-horizon value predictions through the Bellman residual. Our work also extends causal calibration for static treatment-effect prediction (Van Der Laan et al., 2023; Whitehouse et al., 2024) to dynamic, long-horizon MDPs.

van der Laan et al. (2025a) proposed an isotonic post-hoc calibration method for Q-functions analogous to our approach for V-functions. Their goal was to debias global policy-value estimators in Double RL (Kallus & Uehara, 2020; 2022), rather than to improve value prediction itself. They also did not provide theoretical guarantees for the calibration algorithm. Moreover, calibration of a Q-function does not in general imply calibration of the induced value function, so our setting, objective, and guarantees are distinct.

Many practical variants of fitted Q-iteration aim to improve approximation quality or robustness in offline settings, including adversarial or minimax formulations (Uehara et al., 2021; 2020; Jin et al., 2021; Uehara et al., 2023; Xie & Jiang, 2021; Di et al., 2023), boosted methods that iteratively regress Bellman residuals (Tosatto et al., 2017), conservative or pessimistic updates (Kumar et al., 2020; An et al., 2021; Di et al., 2023), operator-regularized variants (massoud Farahmand et al., 2009), structural approaches based on linear or low-rank models (Shah et al., 2020), and distributionally robust methods (Zhou et al., 2021). These approaches aim to stabilize or regularize Bellman updates, often by modifying the effective operator. Our method is complementary: it is a post-hoc calibration step that can in principle be applied on top of any value predictor.

To our knowledge, prior work has not defined or enforced calibration of long-horizon value predictions in RL. Sequential calibration in forecasting and online prediction concerns one-step predictive accuracy and does not impose Bellman consistency. Bellman conformal inference (Yang et al., 2024) constructs calibrated prediction intervals for time-series forecasting rather than calibrating value functions in MDPs. Related work also studies calibrated uncertainty for learned dynamics models (Malik et al., 2019), conformal uncertainty sets in RL (Zhang et al., 2023; Sun et al., 2023), and distributional RL for modeling return distributions (Bellemare et al., 2017; Dabney et al., 2018); these are related in spirit but address different objects.

2. Calibration for Value Functions

2.1. Setup and Notation

We consider an MDP with continuous state space $\mathcal{S}$, discrete action space $\mathcal{A}$, initial state distribution ρ, transition kernel $P(s' \mid s, a)$, reward function $r_0(s, a)$, and discount factor $\gamma \in [0, 1)$. Data are collected under a behavior policy $b_0(a \mid s)$, and each sample consists of a transition (S, A, R, S'), where $(S, A) \sim \rho \times b_0$, $S' \sim P(\cdot \mid S, A)$, and $R = r_0(S, A) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid S, A] = 0$. We ob-

serve a calibration dataset of n i.i.d. transitions, $\mathcal{C}_n := \{(S_i, A_i, R_i, S_i')\}_{i=1}^n$.

Let $\pi(a \mid s)$ denote a target policy and $w_\pi := \pi/b_0$ the corresponding importance ratio. Define $(\pi f)(s) := \sum_{a\in\mathcal{A}} \pi(a \mid s) f(s,a)$ and $(Pv)(s,a) := \mathbb{E}[v(S') \mid S = s, A = a]$. The policy-marginalized quantities are $P_\pi := \pi P$ and $r_{0,\pi} := \pi r_0$, with $\mathcal{T}_\pi(f) := r_{0,\pi} + \gamma P_\pi f$ the associated Bellman operator. A measure μ is *stationary* for a Markov operator P if $\mu P = \mu$.

The *value function* under policy π is the expected discounted return starting in state s and following π:

$$v_0(s) = \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t R_t \,\middle|\, S_0 = s\right].$$

Here $\mathbb{E}_\pi$ denotes expectation over $A \sim \pi(\cdot \mid S)$ and $S' \sim P(\cdot \mid S, A)$. It is the unique bounded fixed point of the Bellman equation (Bellman, 1952; 1966)

$$v_0 = \mathcal{T}_\pi(v_0), \quad \mathcal{T}_\pi(v_0)(s) = r_{0,\pi}(s) + \gamma\,(P_\pi v_0)(s). \quad (1)$$

Let $L^2(\rho)$ denote the space of square-integrable functions of S under ρ, with norm

$$\|f\|_{2,\rho} := \{\mathbb{E}_\rho[f(S)^2]\}^{1/2},$$

and for any measure μ on $\mathcal{S} \times \mathcal{A}$, let $\|f\|_{2,\mu} := \{\int f(s,a)^2\,\mu(ds,da)\}^{1/2}$. Let $\|f\|_{n,S,A}$ and $\|g\|_{n,S'}$ denote the empirical L^2 norms over $\{(S_i, A_i)\}_{i=1}^n$ and $\{S_i'\}_{i=1}^n$, respectively. We write $\lesssim$ for inequalities up to absolute constants, and use $[K] := \{0, 1, \ldots, K\}$.

2.2. Bellman Calibration: A Minimal Requirement

Let $\hat{v}$ denote an estimated value function for the target policy π, fit on data independent of the calibration sample $\mathcal{C}_n$. Our goal is not to relearn the value function from scratch, but to impose a minimal correctness requirement on an existing predictor. We formalize this through *Bellman calibration*, a weaker, score-based analogue of Bellman consistency.

A natural requirement for a value predictor is *Bellman consistency*, namely

$$\hat{v}(S) \approx \mathbb{E}_\pi[R + \gamma \hat{v}(S') \mid S].$$

Equivalently, the Bellman residual $R+\gamma\hat{v}(S')-\hat{v}(S)$ should have conditional mean zero given the full state. In high-dimensional or continuous state spaces, however, this is a strong requirement: enforcing or even checking it generally requires learning the full state-dependent conditional mean of the residual, often under substantial structure on the function class, such as Bellman completeness or linearity.

We instead consider a weaker requirement obtained by coarsening the state space through the predictor itself. Rather than requiring the Bellman residual to vanish conditional on S, we require it to vanish conditional only on the scalar score $\hat{v}(S)$:

$$\mathbb{E}_\pi[R + \gamma\hat{v}(S') - \hat{v}(S) \mid \hat{v}(S)] \approx 0.$$

That is, among states assigned the same predicted long-run return, the prediction should be correct on average. This is the dynamic analogue of post-hoc calibration in supervised prediction, where outcomes are required to be correct on average within level sets of a scalar score (Zadrozny & Elkan, 2002; Niculescu-Mizil & Caruana, 2005; Guo et al., 2017). The benefit is that it replaces a high-dimensional conditional moment restriction with a one-dimensional one, making post-hoc correction feasible without relearning a fully state-dependent value model.

Bellman calibration. For any candidate value function v, define the Bellman-calibration map

$$\Gamma_0(v)(s) := \mathbb{E}_\pi[R + \gamma v(S') \mid v(S) = v(s)]. \quad (2)$$

We say that v is *Bellman calibrated* if

$$v(S) = \Gamma_0(v)(S) \quad \text{almost surely},$$

or equivalently,

$$\mathbb{E}_\pi[R + \gamma v(S') - v(S) \mid v(S)] = 0 \quad \text{almost surely}.$$

Thus, Bellman calibration requires correctness only after grouping states by their predicted value.

We quantify deviations from this condition by the *Bellman calibration error*

$$\mathrm{Cal}_{\ell^2}(v) := \|v - \Gamma_0(v)\|_{2,\rho}. \quad (3)$$

Bellman calibration is necessary but not sufficient. The true value function v_0 is Bellman calibrated. Indeed, by the Bellman equation,

$$v_0(S) = \mathbb{E}_\pi[R + \gamma v_0(S') \mid S] \quad \text{almost surely}.$$

Applying the tower property and conditioning on the coarser variable $v_0(S)$ gives

$$v_0(S) = \mathbb{E}_\pi[R+\gamma v_0(S') \mid v_0(S)] = \Gamma_0(v_0)(S) \quad \text{almost surely}.$$

Thus, any Bellman-consistent value function is automatically Bellman calibrated. The converse need not hold. Because Bellman calibration conditions only on the scalar score $\hat{v}(S)$, it does not rule out substantial heterogeneity in the true value among states with the same predicted score. As a result, even an inaccurate predictor may be Bellman calibrated if its errors average out within score levels (Gupta et al., 2020). Bellman calibration should therefore be viewed

as a minimal reliability requirement, not as a guarantee of overall value accuracy.

Bellman calibration also admits a reward-model interpretation. Define the reward function implied by a value predictor $\hat{v}$ as

$$\hat{r}_{\pi,\hat{v}}(s) := \hat{v}(s) - \gamma P_\pi \hat{v}(s).$$

Then Bellman calibration of $\hat{v}$ is equivalent to

$$\mathbb{E}_\pi[R - \hat{r}_{\pi,\hat{v}}(S) \mid \hat{v}(S)] = 0.$$

That is, the reward model implied by $\hat{v}$ is calibrated for the true reward after grouping states by their predicted long-run return. When $\gamma = 0$, this reduces to ordinary regression calibration (Bella et al., 2010; Roth, 2022).

Strong Bellman calibration. The definition above calibrates $\hat{v}$ only with respect to its own Bellman target, and is therefore a self-consistency condition. A stronger alternative would require calibration with respect to the true value function v_0 itself. Specifically, we say that $\hat{v}$ is *strongly Bellman calibrated* if

$$\hat{v}(S) = \mathbb{E}[v_0(S) \mid \hat{v}(S)] \quad \text{almost surely.}$$

Equivalently, among states with the same predicted value, the prediction agrees on average with the true long-run return under π.

This notion is substantially stronger, since it requires learning the conditional mean of the unknown target $v_0(S)$ given the score $\hat{v}(S)$. In general, this is not directly available from one-step Bellman targets alone and is typically no easier than the underlying value-learning problem itself (Kallus & Uehara, 2020; 2022). For this reason, we focus on the weaker but practically attainable notion of Bellman calibration defined above.

2.3. Bellman calibration and value error

We now clarify what Bellman calibration buys us for value prediction. Since it is only a score-based correctness property, it does not by itself guarantee that $\hat{v}$ is close to v_0. The result below decomposes the error of $\hat{v}$ into a *refinement* term, capturing information lost by reducing the state to the score $\hat{v}(S)$, and a *calibration* term, capturing failure of Bellman calibration within that score class.

Define the class of score-based transformations

$$\mathcal{F}_{\hat{v}} := \{\theta \circ \hat{v} : \theta : \mathbb{R} \to \mathbb{R}\},$$

and let $\Pi_{\hat{v}}$ denote the $L^2(\rho)$ projection onto this class:

$$\Pi_{\hat{v}} f := \arg\min_{g \in \mathcal{F}_{\hat{v}}} \|f - g\|_{2,\rho}.$$

Thus, $\Pi_{\hat{v}} f$ is the best predictor of f that depends on the state only through the scalar score $\hat{v}(S)$.

Let $P_{\pi,\hat{v}} := \Pi_{\hat{v}} P_\pi$ denote the corresponding coarsened transition operator, i.e.,

$$P_{\pi,\hat{v}} f(s) = \mathbb{E}_\pi[f(S') \mid \hat{v}(S) = \hat{v}(s)].$$

We then define the coarsened Bellman operator $\mathcal{T}_{\pi,\hat{v}} q := \Pi_{\hat{v}} \mathcal{T}_\pi q$, and let $\hat{v}_0 \in \mathcal{F}_{\hat{v}}$ denote its fixed point:

$$\hat{v}_0 = \mathcal{T}_{\pi,\hat{v}} \hat{v}_0. \tag{4}$$

The function $\hat{v}_0$ is the oracle Bellman-calibrated predictor in the score class $\mathcal{F}_{\hat{v}}$.

A1 There exists a stationary measure $\mu_{\hat{v}}$ for $P_{\pi,\hat{v}}$.

Under A1, $\mathcal{T}_{\pi,\hat{v}}$ is a γ-contraction on $L^2(\mu_{\hat{v}})$ (Lemma 1 in Appendix B). In the on-policy setting, where ρ is stationary for P_π, A1 holds with $\mu_{\hat{v}} := \rho$ by Lemma 2 in Appendix B.

Theorem 1 (Calibration–Refinement Bound)**.** *Under A1,*

$$\|\hat{v} - v_0\|_{2,\mu_{\hat{v}}} \;\le\; \frac{1}{1-\gamma}\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}} \;+\; \|\hat{v} - \hat{v}_0\|_{2,\mu_{\hat{v}}}.$$

Theorem 1 separates two sources of value error. The *refinement error* $\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}$ measures the information lost by compressing the state to the scalar score $\hat{v}(S)$, and vanishes only if v_0 is recoverable from that score. The *calibration error* $\|\hat{v} - \hat{v}_0\|_{2,\mu_{\hat{v}}}$ measures how far $\hat{v}$ is from the Bellman-calibrated oracle within the same score class. Thus, Bellman calibration cannot recover information lost by the one-dimensional reduction, but it can remove the additional error due to lack of Bellman self-consistency within that score class. This mirrors classical calibration–refinement decompositions in classification and regression (Murphy, 1973; DeGroot & Fienberg, 1983; Van Der Laan et al., 2023; Whitehouse et al., 2024).

If $\hat{v}$ is Bellman calibrated, then $\hat{v} = \hat{v}_0$ and the calibration term vanishes. The remaining error is then exactly the irreducible refinement error associated with the score $\hat{v}(S)$, up to the factor $(1-\gamma)^{-1}$. Thus, once $\hat{v}$ is calibrated, no further post-hoc transformation of the form $\theta \circ \hat{v}$ can improve the bound without changing the underlying score itself.

On-policy vs off-policy. Under on-policy sampling from a stationary distribution, the behavior distribution ρ is stationary for P_π, and hence also for the coarsened kernel $P_{\pi,\hat{v}}$. Therefore we may take $\mu_{\hat{v}} = \rho$, so the decomposition is measured in the same $L^2(\rho)$ norm:

$$\|\hat{v} - v_0\|_{2,\rho} \;\le\; \frac{1}{1-\gamma} \min_{\theta:\mathbb{R}\to\mathbb{R}} \|\theta \circ \hat{v} - v_0\|_{2,\rho} + \|\hat{v} - \hat{v}_0\|_{2,\rho}.$$

In this case, Bellman calibration affects only the second term: it makes $\hat{v}$ as accurate as possible among predictors that depend on the state only through the score $\hat{v}(S)$.

Under off-policy sampling, this simplification breaks down because $\Pi_{\hat{v}}$ is defined in the behavior norm $L^2(\rho)$, whereas Bellman contraction is governed by the stationary norm $L^2(\mu_{\hat{v}})$ of the coarsened kernel. This norm mismatch is the main difficulty in the off-policy setting (Patterson et al., 2022; van der Laan & Kallus, 2025), and is formalized later in Theorem 4.

2.4. A Doubly Robust Bellman Target

Off-policy prediction requires estimating $\mathcal{T}_\pi(v)$ from data generated under the behavior policy b_0. Importance weighting with $w_\pi = \pi/b_0$ corrects this distribution shift, but can be unstable when the weights are poorly estimated. To obtain a more stable target, we use a doubly robust Bellman pseudo-outcome that combines importance weighting with estimates of the reward and transition functions. This target is unbiased if either the weights or the reward–transition model is correctly specified.

Let $\widehat{w}_\pi$ estimate $w_\pi := \pi/b_0$, let $\hat{r}$ estimate the reward function r_0, and let $\widehat{P}$ estimate the Markov operator P. For any function v, define the doubly robust fitted Bellman target $\widehat{\mathcal{T}}_\pi(v)(S, A, R, S')$ as

$$(\pi \hat{q}_v)(S) + \widehat{w}_\pi(A \mid S)\big[R + \gamma v(S') - \hat{q}_v(S, A)\big],$$
$$\text{where } q_v := r_0 + \gamma P v \text{ and } \hat{q}_v := \hat{r} + \gamma \widehat{P} v.$$

When $\gamma = 0$, this reduces to the standard doubly robust pseudo-outcome from static treatment-effect estimation (Rubin & van der Laan, 2006; Van Der Laan et al., 2023; Kennedy, 2023).

Recall that $(b_0 f)(s) := \sum_{a \in \mathcal{A}} b_0(a \mid s)\, f(s, a)$ and define the denoised Bellman operator

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) \;:=\; \mathbb{E}\Big[\widehat{\mathcal{T}}_\pi(v)(S, A, R, S') \,\Big|\, S = s\Big].$$

Theorem 2 (Doubly robust errors)**.** *For any v,*

$$\widehat{\mathcal{T}}_{0,\pi}(v) - \mathcal{T}_\pi(v) = b_0\{(w_\pi - \widehat{w}_\pi)(\hat{q}_v - q_v)\}.$$

The fitted Bellman target is *doubly robust*: it is unbiased whenever $\widehat{w}_\pi = w_\pi$ or $\hat{q}_v = q_v$ (for example, if $\widehat{r} = r_0$ and $\widehat{P} = P$). The ratio w_π is known whenever the behavior policy b_0 is known, and in particular when $b_0 = \pi$, in which case $w_\pi \equiv 1$. Then the pseudo-outcome remains valid even with trivial nuisance estimates $\widehat{r} = 0$ and $\widehat{P} v = 0$, reducing to the standard importance-weighted target

$$\widehat{\mathcal{T}}_\pi(v)(S, A, R, S') = w_\pi(A \mid S)\big(R + \gamma v(S')\big). \quad (5)$$

Conversely, in settings such as robotics or simulation where the transition kernel is known, so that $\widehat{P} = P$, the bias reduces to $b_0\{(\widehat{w}_\pi - w_\pi)(\widehat{r} - r_0)\}$. Thus doubly robustness then applies to the weight and reward estimators alone.

Algorithm 1 Iterated Bellman Calibration

input Value predictor $\hat{v}$, calibration data $\mathcal{C}_n$, fitted Bellman operator $\widehat{\mathcal{T}}_\pi$, calibrator class $\mathcal{F}$, iterations K
1: Initialize $\hat{v}^{(0)} \leftarrow \hat{v}$
2: **for** $k = 0, \ldots, K-1$ **do**
`# Construct Bellman target`
3: $\hat{\chi}_i^{(k)} \leftarrow \widehat{\mathcal{T}}_\pi(\hat{v}^{(k)})(S_i, A_i, R_i, S_i')$
`# Fit 1D calibrator`
4: $\theta_n^{(k+1)} \leftarrow \arg\min_{\theta \in \mathcal{F}} \sum_{i=1}^n (\hat{\chi}_i^{(k)} - \theta(\hat{v}(S_i)))^2$
`# Update predictor`
5: $\hat{v}^{(k+1)} \leftarrow \theta_n^{(k+1)} \circ \hat{v}$
6: **end for**
output $\hat{v}^{(K)}$

3. Algorithms and Theory

3.1. Iterated Bellman Calibration Algorithm

Algorithm 1 introduces a simple post-hoc calibration method, ***Iterated Bellman Calibration***, which transforms an initial value predictor $\hat{v}$ into a calibrated predictor of the form

$$\hat{v}^{\text{cal}} := \theta_n \circ \hat{v},$$

where the calibrator $\theta_n : \mathbb{R} \to \mathbb{R}$ is learned from the calibration data $\mathcal{C}_n$. At a high level, the method performs fitted Bellman iteration on the one-dimensional score $\hat{v}(S)$: at each step, it regresses doubly robust Bellman targets $\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(k)})(S_i, A_i, R_i, S_i')\}_{i=1}^n$ onto the initial scores $\{\hat{v}(S_i)\}_{i=1}^n$ using a regression class $\mathcal{F}$, and sets $\hat{v}^{(k+1)} = \theta_n^{(k+1)} \circ \hat{v}$. Relative to standard fitted value iteration (Munos, 2005; Munos & Szepesvári, 2008), the key restriction is that every iterate remains in the score class $\{\theta \circ \hat{v} : \theta \in \mathcal{F}\}$, thereby avoiding high-dimensional function approximation. Since the finite-iteration error decays geometrically, only a small number of iterations is needed in practice.

Heuristically, $\hat{v}^{\text{cal}}$ is calibrated because the iteration targets the fixed point $\hat{v}_0$ of the coarsened Bellman operator in (4), which is Bellman calibrated. In finite samples, recovering this fixed point exactly would typically require a realizability condition—namely, that $\hat{v}_0$ belongs to the score class $\{\theta \circ \hat{v} : \theta \in \mathcal{F}\}$—together with Bellman completeness for the fitted regression step. The methods introduced next avoid these strong requirements.

We study assumption-light instantiations of Algorithm 1 based on histogram regression (Stone, 1977) and isotonic regression (Barlow & Brunk, 1972). These extend standard calibration methods—namely, histogram binning (Zadrozny & Elkan, 2001; 2002; Gupta & Ramdas, 2021) and isotonic calibration (Zadrozny & Elkan, 2002; Niculescu-Mizil & Caruana, 2005; Van Der Laan et al., 2023; van der Laan et al., 2024b; van der Laan & Alaa, 2025)—from static prediction problems to the dynamic setting, and admit finite-

sample Bellman calibration guarantees.

3.2. Histogram Calibration

Histogram calibration learns a piecewise-constant post-hoc transform by grouping states with similar initial predictions. Specifically, we partition the scores $\{\hat{v}(S_i)\}_{i=1}^n$ into B ordered bins $\{I_1, \dots, I_B\}$, formed either by empirical quantiles (*equal-mass* binning) or by uniform discretization of the prediction range (*equal-width* binning). We then apply Algorithm 1 with calibrator class $\mathcal{F}_B := \mathrm{span}\{\mathbf{1}_{I_b} : b \in [B]\}$, so that each iterate is constant within bins. The bins may be data-adaptive, provided the number of bins B is deterministic, possibly growing with n.

Operationally, histogram calibration updates each bin value by averaging the fitted Bellman targets within that bin. At iteration K, the calibrator is piecewise constant:

$$\theta_n^{(K)}(t) = n_b^{-1} \sum_{i:\, \hat{v}(S_i) \in I_b} \hat{\chi}_i^{(K)} \quad \text{for } t \in I_b,$$

where $n_b := \sum_{i=1}^n \mathbf{1}\{\hat{v}(S_i) \in I_b\}$. Thus, each update simply replaces the value on a bin by the average Bellman target over that bin.

This binwise update yields the empirical relation

$$\hat{v}^{(K)}(s) = \mathbb{E}_n\left[\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(S, A, R, S') \,\middle|\, \hat{v}^{(K)}(S) = \hat{v}^{(K)}(s)\right],$$

where $\mathbb{E}_n$ denotes empirical conditional expectation over the calibration sample $\mathcal{C}_n$. Hence, when the iteration stabilizes ($\hat{v}^{(K)} \approx \hat{v}^{(K-1)}$), the output is approximately an empirical fixed point of the coarsened Bellman equation, and therefore approximately Bellman calibrated.

Calibration error. We next derive finite-sample, finite-iteration bounds on the Bellman calibration error (3).

The following technical conditions simplify the analysis and could be relaxed with additional work.

C1 (*Boundedness*) R, r, $\widehat{r}$, w_π, $\widehat{w}_\pi$, and $\hat{v}$ are uniformly bounded by a constant $M \in (0, \infty)$.

C2 (*Sample splitting*) The estimators $\widehat{r}$, $\widehat{w}_\pi$, $\widehat{P}$, and $\hat{v}$ are obtained from data independent of $\mathcal{C}_n$.

C3 ($\widehat{P}$ *is L–Lipschitz*) For all $f, g \in \mathcal{F}_B$ with $\|f\|_\infty, \|g\|_\infty \le M$, $\|\widehat{P}(f \circ \hat{v}) - \widehat{P}(g \circ \hat{v})\|_{n,A,S} \le L\,\|(f - g) \circ \hat{v}\|_{n,S'}$.

Condition C1 can be relaxed using weight clipping together with appropriate tail conditions on the density ratio and outcomes, following variance-control ideas from importance-weighted RL (Hallak et al., 2016; Han & Sung, 2019; Mehrabi & Wager, 2024) and from heavy-tailed regression. Condition C2 can similarly be relaxed by standard cross-fitting arguments (Van Der Laan et al., 2023; Foster & Syrgkanis, 2023). For classification with $\gamma = 0$, Gupta & Ramdas (2021) show that outcome-agnostic histogram binning allows $\hat{v}$ to be fit on the same data used for calibration. Condition C3 simplifies the analysis by controlling the metric entropy of $\{\widehat{P} f \circ \hat{v} : f \in \mathcal{F}_B\}$ through that of $\mathcal{F}_B$. It holds trivially when $\widehat{P} = 0$, as in the importance-weighted target (5), and it holds with $L = 1$ when $\widehat{P}$ is a discrete nonparametric MLE. When $\widehat{P} = P$, the condition holds at the population level and, under mild regularity conditions, also empirically with high probability.

Theorem 3 (Calibration Error for Histogram Binning)**.** *Assume C1-C3. Then, there exists a $C \in (0, \infty)$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,*

$$\begin{aligned}\mathrm{Cal}_{\ell^2}(\hat{v}^{(K)}) \;\le\; & C\left(\sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\delta)}{n}}\right) \\ & + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho} \\ & + \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\|_{2,\rho}.\end{aligned}$$

The first two terms are the statistical error of one-dimensional histogram calibration: they are exactly the rates one would obtain if the calibration target were observed directly, matching classical bounds for histogram binning (Gupta et al., 2020; Gupta & Ramdas, 2021; van der Laan & Alaa, 2025). The third term is the doubly robust nuisance error from Theorem 2. The final term is the finite-iteration error; under mild conditions it decays geometrically in K (Appendix G.4), in line with fitted value iteration theory (Munos & Szepesvári, 2008).

Notably, the leading calibration terms do not depend on the discount factor γ except through the finite-iteration term. This reflects the fact that Bellman calibration is a one-step conditional prediction problem: unlike value estimation, its statistical error does not accumulate over future timesteps. Thus calibration is substantially easier to control statistically. By Theorem 1, however, the extent to which calibration translates into value accuracy still deteriorates as γ increases and distributional coverage weakens, as we now formalize.

Estimation error. The next theorem shows how Bellman calibration affects the value-estimation error of the original predictor $\hat{v}$. In particular, post-hoc calibration does not introduce any error beyond that already incurred by restricting attention to the histogram score class, and may reduce error due to Bellman inconsistency. This is especially relevant under model misspecification, where calibration can improve self-consistency without relearning a high-dimensional value function.

Define the $L^2(\rho)$ projection onto the histogram score class

$$\Pi_{\hat{v},B} q := \arg\min_{\theta \circ \hat{v} : \theta \in \mathcal{F}_B} \|q - \theta \circ \hat{v}\|_{2,\rho}.$$

Let $\hat{v}_{0,B} \in \{\theta \circ \hat{v} : \theta \in \mathcal{F}_B\}$ denote the unique fixed point of the corresponding coarsened Bellman operator

$$\hat{v}_{0,B} = \mathcal{T}_{\pi,\hat{v},B}\hat{v}_{0,B}, \qquad \mathcal{T}_{\pi,\hat{v},B}q := \Pi_{\hat{v},B}\mathcal{T}_\pi q.$$

The associated coarsened transition operator is $P_{\pi,\hat{v},B} := \Pi_{\hat{v},B}P_\pi$, which satisfies

$$P_{\pi,\hat{v},B}f(s) = \mathbb{E}_\pi\left[f(S') \mid \hat{v}(S) \in I_{\hat{b}(s)}\right],$$

where $\hat{b}(s) = b$ whenever $\hat{v}(s) \in I_b$.

C4 (*Stationary coverage*) There exists a stationary measure $\mu_{\hat{v},B}$ for $P_{\pi,\hat{v},B}$ and $\hat{\kappa}_B \in (0,\infty)$ such that $\|h\|_{2,\mu_{\hat{v},B}} \le \hat{\kappa}_B \|h\|_{2,\rho}$ for all $h : \mathcal{S} \to \mathbb{R}$.

Theorem 4 (Estimation Error for Histogram Binning)**.** *Assume C1-C4. Then, there exists a $C \in (0,\infty)$ such that, for any $K \in \mathbb{N}$, with probability at least $1-\delta$,*

$$\begin{aligned}
&\|\hat{v}^{(K)} - v_0\|_{2,\mu_{\hat{v},B}} \\
&\quad \le \gamma^K \|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{\hat{\kappa}_B}{1-\gamma} \min_{\theta\in\mathcal{F}_B} \|\theta\circ\hat{v} - v_0\|_{2,\rho} \\
&\quad + \frac{\hat{\kappa}_B C}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\frac{n}{B}} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\hat{\kappa}_B}{1-\gamma} \max_{j\in[K]} \left\|(\hat{w}_\pi - w_\pi)(\hat{q}_{\hat{v}^{(j)}} - q_{\hat{v}^{(j)}})\right\|_{2,\rho}.
\end{aligned}$$

Remark. The proof also yields the same bound for $\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}}$, but without the approximation term $\min_{\theta\in\mathcal{F}_B} \|\theta\circ\hat{v} - v_0\|_{2,\rho}$. Thus, the algorithm converges rapidly to the coarsened fixed point $\hat{v}_{0,B}$ even if the initial predictor $\hat{v}$ is poor.

Proof technique. Theorem 4 does requires neither realizability nor Bellman completeness for the original value class. The key point is that Bellman completeness holds automatically for the coarsened Bellman operator $\mathcal{T}_{\pi,\hat{v},B}$. Indeed, defining $\mathcal{F}_{B,\hat{v}} := \{\theta\circ\hat{v} : \theta \in \mathcal{F}_B\}$, we have $P_{\pi,\hat{v},B}f \in \mathcal{F}_{B,\hat{v}}$ for every $f \in \mathcal{F}_{B,\hat{v}}$, since $P_{\pi,\hat{v},B}$ averages over bins of $\hat{v}$ and therefore produces a function that is constant on those bins. Hence $\mathcal{T}_{\pi,\hat{v},B}f \in \mathcal{F}_{B,\hat{v}}$ as well.

The proof therefore proceeds in two steps: first, it shows that $\hat{v}^{(K)}$ converges to the fixed point $\hat{v}_{0,B}$ of the coarsened operator; second, it controls the gap $\hat{v}_{0,B} - v_0$ via a modification of Theorem 1. A related use of discretization to induce piecewise-constant value classes and avoid Bellman completeness assumptions appears in Xie & Jiang (2021). That work discretizes a high-dimensional state space and targets Bellman consistency, whereas we bin the scalar score $\hat{v}(S)$, yielding a simpler discretization tailored to Bellman calibration.

Discussion. When the importance weights w_π are known and K is sufficiently large, Theorem 3 implies that the calibration error vanishes at rate $\sqrt{(B/n)\log(n/B)}$. By contrast, Theorem 4 shows that the estimation error contains two terms: an approximation term

$$\hat{\kappa}_B(1-\gamma)^{-1} \min_{\theta\in\mathcal{F}_B} \|\theta\circ\hat{v} - v_0\|_{2,\rho},$$

reflecting the loss from restricting attention to binned transformations of $\hat{v}$, and the same stochastic term $\sqrt{(B/n)\log(n/B)}$. Thus the choice of B induces the usual bias–variance tradeoff: small B yields a coarse calibrator and potentially large approximation error, while large B increases variance by leaving too few samples per bin.

To preserve the predictive structure of $\hat{v}$, the binning should be fine enough that $\mathcal{F}_B$ can approximate the identity map well. Indeed,

$$\min_{\theta\in\mathcal{F}_B} \|\theta\circ\hat{v} - v_0\|_{2,\rho} \;\le\; \|\hat{v} - v_0\|_{2,\rho} + \min_{\theta\in\mathcal{F}_B} \|\theta\circ\hat{v} - \hat{v}\|_{2,\rho}. \tag{6}$$

Thus, if $\mathcal{F}_B$ approximates the identity map accurately, the best binned transformation of $\hat{v}$ is nearly as accurate as $\hat{v}$ itself.

For equal-mass binning, choosing $B \asymp n^{1/3}$ yields the worst-case rate (Györfi et al., 2002)

$$\|\hat{v}^{(K)} - v_0\|_{2,\mu_{\hat{v},B}} \;\le\; \frac{\hat{\kappa}_B}{1-\gamma}\|\hat{v} - v_0\|_{2,\rho} + O_p((\log n/n)^{1/3}),$$

and under the same scaling the calibration error also vanishes at rate $(\log n/n)^{1/3}$. This suggests that choosing B to optimize predictive accuracy, for example by cross-validation, will typically also yield good calibration.

Role of the stationary coverage constant. The stationary coverage constant $\hat{\kappa}_B$ in C4 quantifies how well the behavior (sampling) norm controls the stationary norm $L^2(\mu_{\hat{v},B})$. Smaller $\hat{\kappa}_B$ therefore yield sharper guarantees. In the most favorable case, on-policy sampling from the stationary distribution of (P,π) aligns these norms and gives $\hat{\kappa}_B = 1$. As the data distribution moves farther from stationarity, $\hat{\kappa}_B$ can grow, reflecting weaker stationary overlap and looser error bounds. Thus, Bellman calibration is easiest in the on-policy stationary regime and becomes harder as off-policy mismatch increases.

3.3. Isotonic Calibration

Histogram calibration requires choosing the number of bins B to balance bias and variance. As a tuning-light alternative, we consider *isotonic iterated Bellman calibration*, obtained by taking $\mathcal{F} := \mathcal{F}_{\text{iso}}$ in Algorithm 1, where $\mathcal{F}_{\text{iso}}$ is the class of nondecreasing functions on $\mathbb{R}$. Each update then performs isotonic regression of the fitted Bellman targets onto the initial scores $\hat{v}(S_i)$, yielding a monotone

calibrator $\theta_n^{(k)} \in \mathcal{F}_{\text{iso}}$. This step can be computed in near-linear time via the pool-adjacent-violators algorithm (Best & Chakravarti, 1990).

Like histogram calibration, isotonic calibration pools nearby score levels and replaces them by averaged Bellman targets, but it does so using a data-adaptive partition determined by the monotonicity constraint (van der Laan & Alaa, 2025). This provides regularization without requiring a choice of bin number. Moreover, because the identity map is monotone, isotonic calibration is less likely to distort an already reliable predictor.

C5 (*Finite variation of the calibrated target*) There exists $V \in (0, \infty)$ such that, almost surely, the function $t \mapsto \mathbb{E}[\mathcal{T}_\pi(\hat{v}^{(K)})(S) \mid \hat{v}(S) = t, \mathcal{C}_n]$ has total variation at most V.

Condition C5 is also assumed in Van Der Laan et al. (2023) and is needed to control the calibration function $\Gamma_0(\hat{v})$; for histogram binning, this complexity is governed by B.

Theorem 5 (Calibration Error for Isotonic Calibration)**.** *Assume C1-C5 for $\mathcal{F}_{iso}$. Then, there exists a $C \in (0, \infty)$ such that, for any $K \in \mathbb{N}$, with probability at least $1 - \delta$,*

$$\begin{aligned}\mathrm{Cal}_{\ell^2}(\hat{v}^{(K)}) \;\le\; C\left(n^{-1/3} + \sqrt{\frac{\log(1/\delta)}{n}}\right) \\ + \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho} \\ + \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\|_{2,\rho}.\end{aligned}$$

The isotonic case parallels the histogram analysis, but uses the data-adaptive partition induced by the pool-adjacent-violators algorithm. Importantly, we do not assume the calibration target is monotone; monotonicity serves only as a regularization device for constructing an adaptive partition of $\hat{v}$. The resulting $n^{-1/3}$ rate matches the classical isotonic regression rate (Chatterjee et al., 2013; van der Laan et al., 2024a; van der Laan & Alaa, 2025). Thus, isotonic calibration achieves the same calibration rate as histogram binning with $B \asymp n^{1/3}$, without requiring the number of bins to be chosen in advance.

3.4. Hybrid Isotonic–Histogram Calibration

While isotonic iterated Bellman calibration admits finite-sample Bellman calibration guarantees, extending the analysis to obtain an analogue of Theorem 4 is more challenging. The difficulty is that the induced partition changes across iterations. Although isotonic regression can be viewed as histogram regression on a data-adaptive partition, the analysis in Section 3.2 relies on that partition remaining fixed. By contrast, applying Algorithm 1 with $\mathcal{F} = \mathcal{F}_{\text{iso}}$ yields a new data-dependent partition at each step, so the iterates no longer correspond to a single coarsened Bellman operator. This prevents a direct extension of the histogram argument, although the procedure may still be stable in practice when the learned partitions vary little across iterations.

Algorithm 2 Iso–Hist Iterated Bellman Calibration

input Value predictor $\hat{v}$, calibration data $\mathcal{C}_n$, fitted Bellman operator $\widehat{\mathcal{T}}_\pi$, iterations K

Stage 1: fit outcome-adaptive partition
  $\hat{\chi}_i \leftarrow \widehat{\mathcal{T}}_\pi(\hat{v})(S_i, A_i, R_i, S_i')$
  $\theta_{n,\text{iso}} \leftarrow \arg\min_{\theta \in \mathcal{F}_{\text{iso}}} \sum_{i=1}^n (\hat{\chi}_i - \theta(\hat{v}(S_i)))^2$
  Extract bins $\hat{I}_1, \ldots, \hat{I}_{\hat{B}}$ from flat regions of $\theta_{n,\text{iso}}$
  Define $\widehat{\mathcal{F}}_{\hat{B}} := \{\theta : \theta \text{ is constant on each } \hat{I}_b\}$
Stage 2: histogram calibration
  Apply Algorithm 1 with class $\widehat{\mathcal{F}}_{\hat{B}}$ to obtain $\hat{v}^{(K)}$

output $\hat{v}^{(K)}$

To obtain formal estimation guarantees while retaining adaptive binning, we introduce a hybrid isotonic–histogram method (Algorithm 2). The idea is simple: use a single isotonic regression step to learn a data-adaptive partition of the range of $\hat{v}$, then hold this partition fixed and run histogram-based Bellman calibration on top of it. Concretely, we first regress the initial Bellman target on $\hat{v}$ using isotonic regression to obtain an adaptive partition, and then apply Algorithm 1 over this fixed partition for all subsequent iterations. This combines the adaptivity of isotonic calibration with the fixed coarsened-operator structure needed for the histogram analysis. Related uses of isotonic regression to construct adaptive partitions appear in conformal prediction (Nouretdinov et al., 2018; van der Laan & Alaa, 2024).

Given a deterministic bound on the number of bins produced by the isotonic regression step, the guarantees of Section 3.2 for histogram binning apply directly. More generally, the following result extends to any data-adaptive histogram binning method, such as regression trees.

Theorem 6 (Calibration and Estimation Error for Alg. 2)**.** *Let $\hat{v}^{(K)}$ denote the output of Algorithm 2. Suppose the isotonic regression step produces at most $\hat{B}$ bins, where $\hat{B} \le B_n$ for some deterministic sequence B_n with probability at least $1 - \delta$. Assume C1-C5 with $B = B_n$. Then, with probability at least $1 - 2\delta$, the guarantees in Theorems 3 and 4 hold for $\hat{v}^{(K)}$ with $B := B_n$.*

Existing isotonic regression theory suggests that the effective number of induced bins satisfies $\hat{B} = O(n^{1/3})$ when the calibration function is continuously differentiable (Deng et al., 2021); this rate can also be enforced by directly constraining the number of constant segments (van der Laan & Alaa, 2024). Consequently, the B-dependent stochastic term in Theorem 4 scales as $O_p(n^{-1/3})$, consistent with

Theorem 5.

For the approximation term in Theorem 4 to be no larger than the initial error $\|\hat{v}-v_0\|_{2,\rho}$ up to constants, the induced histogram class must also approximate the identity map on the range of $\hat{v}$. That is, we require $\min_{\theta\in\widehat{\mathcal{F}}_{\widehat{B}}} \|\theta\circ\hat{v}-\hat{v}\| \approx 0$, so that step functions in $\widehat{\mathcal{F}}_{\widehat{B}}$ nearly preserve the original predictions. This condition is plausible when $\hat{v}$ is already close to v_0, in which case isotonic regression should produce a near-identity transformation. By contrast, when $\hat{v}$ is substantially miscalibrated, isotonic regression may induce a coarser partition that pools regions of poor prediction into a common calibrated value.

4. Illustrative Experiments

We use synthetic data to compare the proposed calibration methods in a controlled setting. Our goal is not to provide a comprehensive benchmark or to argue that calibration uniformly outperforms strong uncalibrated baselines. That question is already well studied in supervised learning (Zadrozny & Elkan, 2002; Guo et al., 2017), and a broad empirical evaluation in reinforcement learning is beyond the scope of this paper.

Synthetic environment. We evaluate all calibration methods with $K = 1000$ in a synthetic subscription MDP designed to mimic monthly retention and revenue dynamics (Appendix A). We simulate offline data from $n_{\text{cust}} = 50{,}000$ customers over $T = 24$ months with $\gamma = 0.99$ under a fixed heuristic behavior policy, and evaluate off-policy value estimation for a deterministic aggressive target policy, with importance weights in $(0.1, 0.99)$. Performance is measured by $(1-\gamma)$-scaled RMSE on initial states, using Monte Carlo rollouts as ground truth and averaging over 50 runs. In all tables, calibrated estimators are compared to a *Full Raw* baseline, defined as the same underlying estimator retrained on the full dataset without calibration. Additional experimental settings are reported in Appendix A. For quantile binning, we choose B so that each bin contains at least 200 observations.

4.1. Results

Table 1 reports off-policy value estimation error under sample splitting, where a proportion p_{tr} of the data is used to fit the base estimator and the remainder is reserved for calibration. This isolates the behavior of the proposed calibration procedures across different train-calibration splits, rather than comparing calibration to retraining on the full dataset. Performance depends strongly on how data is allocated between fitting and calibration. When $p_{\text{tr}} = .5$, calibration generally does not improve on the full raw baseline, likely because the base estimator is trained on too little data. In contrast, for $p_{\text{tr}} = .75$ and $p_{\text{tr}} = .9$, isotonic and hybrid isotonic calibration consistently improve the linear and neural estimators. The boosted estimator, however, remains strongest without further calibration; as a tree- or histogram-based method, it appears reasonably well calibrated already in these experiments. Across these settings, the hybrid method performs similarly to plain isotonic calibration, with no systematic advantage here.

p_{tr}	Model	Full Raw	Iso	Histogram	Hybrid Iso
.5	Boost	**.34±.03**	.40±.06	.40±.06	.40±.06
	Lin	.63±.02	.62±.03	**.62±.03**	.62±.03
	Neur	**.32±.05**	.35±.05	.35±.05	.35±.05
.75	Boost	**.34±.03**	.37±.05	.37±.05	.37±.05
	Lin	.63±.02	**.61±.03**	.61±.03	.61±.03
	Neur	.33±.07	**.31±.04**	.31±.04	.31±.04
.9	Boost	**.34±.03**	.37±.04	.37±.03	.37±.04
	Lin	.63±.02	**.61±.04**	.61±.04	.61±.04
	Neur	.33±.08	**.31±.04**	.33±.04	.31±.04

Table 1. Main results for $n = 50{,}000$ across training proportions p_{tr} under sample splitting. Lower is better.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.314±0.06	**0.494±0.05**	0.494±0.04	0.497±0.05
25	0.492±0.12	**0.386±0.05**	0.391±0.05	0.388±0.05
50	**0.301**±0.04	0.324±0.03	0.335±0.03	0.324±0.03
100	0.325±0.08	**0.312±0.04**	0.325±0.04	0.312±0.04

Table 2. Neural snapshot performance with sample splitting for $n = 50{,}000$ and training proportion $p_{\text{train}} = 0.9$. Lower is better.

Table 2 examines neural snapshots during training for $p_{\text{tr}} = .9$. Calibration yields large improvements at early iterations, substantially reducing error at iterations 10 and 25 when the raw estimator is still poorly calibrated. At iteration 50, however, the raw snapshot is best, suggesting that calibration can occasionally over-correct a strong intermediate iterate. By iteration 100, isotonic and hybrid calibration again achieve the lowest error. Overall, these results suggest that Bellman calibration is most useful for under-trained or moderately misspecified estimators, with smaller and less uniform gains once the base estimator is already well tuned.

5. Conclusion and Future Work

We introduced *Iterated Bellman Calibration*, a post-hoc method for improving the reliability of off-policy value predictors in infinite-horizon MDPs. Our analysis establishes finite-sample guarantees and convergence to a coarsened Bellman fixed point under weak assumptions, without Bellman completeness or realizability. It also highlights the role of sampling geometry: calibration is most favorable under on-policy sampling from the target policy's stationary distribution, where the Bellman operator and the calibration

step are naturally aligned, and becomes more difficult as off-policy distribution shift grows.

These findings suggest several directions for future work. One is to reduce off-policy mismatch by calibrating under distributions closer to the target stationary distribution, for example using stationary-density-ratio weighting or related emphatic-style corrections (van der Laan & Kallus, 2025; Mahmood et al., 2015; Patterson et al., 2022). Another is to move beyond binning-based calibrators. While histogram and isotonic methods are simple, assumption-light, and theoretically convenient, they do not extrapolate beyond the support of the calibration data. Under stronger distribution shift, it may therefore be useful to study smoother second-stage calibrators, such as temperature-scaling-type corrections (Platt et al., 1999; Jiang et al., 2011; Guo et al., 2017) or other low-dimensional parametric calibration maps. More broadly, an important next step is a systematic empirical study of calibration in RL, covering on-policy and off-policy regimes and comparing simple binning methods, weighting-based corrections, and smoother learned calibrators.

Impact Statement.

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Experimental results and info

A.1. Synthetic CRM environment

We evaluate all methods in a synthetic customer-relationship-management (CRM) Markov decision process (MDP) that mimics monthly retention and revenue dynamics for a subscription service. The state $s_t \in \mathbb{R}^6$ encodes tenure in months, engagement, fatigue, value segment, price sensitivity, and an indicator of whether the customer is active. At each time t, the agent selects one of three actions: no promotion, light promotion, or strong promotion. Rewards correspond to monthly revenue after discounts and are zero after churn.

Transitions capture key CRM effects. Churn probability depends on tenure, engagement, fatigue, and action via a logistic model; visit probability depends on engagement, fatigue, and action; and revenue conditional on visit scales a baseline value segment by an action-dependent uplift and a price-sensitivity effect, with log-normal noise. Engagement decays over time but is boosted by successful promotions, while fatigue increases with promotion intensity and decays slowly otherwise. Once a customer churns or reaches a maximum tenure of 60 months, the process enters an absorbing state.

We simulate offline datasets of $n_{\text{cust}} = 50{,}000$ customers over a horizon of $T = 24$ months with discount factor $\gamma = 0.99$. Behavior data are generated by a fixed heuristic policy that sends light promotions by default, suppresses promotions for highly engaged and fatigued customers, and occasionally sends strong promotions to low-engagement, high-value customers. We evaluate off-policy value estimation for a deterministic *aggressive* revenue-seeking target policy that sends strong promotions to low-engagement, high-sensitivity customers and avoids promotions for highly engaged, high-value customers.

Ground-truth values $V^\pi(s_0)$ are estimated via Monte Carlo rollouts in the same environment, and we report $(1-\gamma)$-scaled RMSE between the estimated and Monte Carlo values at initial states, averaged across 50 independent runs.

Base estimators. We consider three base estimators: boosted FQE, linear FVI, and neural FVI. The linear model uses a hand-crafted semi-structured feature map consisting of the six raw state variables, normalized and log-transformed tenure, squared terms for engagement, fatigue, value segment, and price sensitivity, three state-only interaction terms, and additive action one-hot indicators, with no state-action interactions. The neural model takes the raw state concatenated with the action as input and uses a feedforward ReLU network with three hidden layers of width 256. Both FVI estimators are trained by iterating one-step Bellman regressions: the linear model uses ridge-regularized least squares, while the neural model is trained for 100 outer iterations using Adam with learning rate 0.005, one epoch per iteration, and mini-batch size 2000. We do not tune the number of neural iterations, since part of the goal is to illustrate a realistic setting in which a base estimator remains poorly calibrated. Selected neural iterates are saved for the snapshot comparisons. Both FVI estimators use an internal 80/20 train-validation split. The boosted baseline is implemented as a LightGBM FQE model trained on the concatenated state-action input. Under sample splitting, base estimators are fit on the training subset and Bellman calibrators on the held-out calibration subset; we also retrain each base estimator on the full dataset to obtain the *Full Raw* baselines.

For the calibration methods, we use a large fixed number of iterations (1000) and verify convergence by visual inspection.

A.2. Calibrating on full data: no sample splitting

Table 3 reports the main results when calibration is applied on the full dataset, without sample splitting. Table 4 complements this comparison by showing neural FVI snapshot performance across iterations in the same no-splitting setting.

Model	Full Raw	Iso	Histogram	Hybrid Iso
Boost	**0.34±0.03**	0.38±0.07	0.39±0.08	0.38±0.07
Lin	0.63±0.02	**0.62±0.02**	0.62±0.02	0.62±0.02
Neur	0.33±0.08	0.30±0.04	0.30±0.04	**0.30±0.04**

Table 3. Main results for $n = 50{,}000$ without sample splitting, where calibration is performed on the full dataset. Lower is better.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.293±0.07	0.478±0.05	0.480±0.05	**0.478±0.05**
25	0.452±0.09	**0.347±0.05**	0.347±0.05	0.347±0.05
50	0.324±0.06	0.298±0.04	0.298±0.04	**0.298±0.04**
100	0.325±0.08	0.300±0.04	0.300±0.04	**0.300±0.04**

Table 4. Neural snapshot performance without sample splitting for $n = 50{,}000$ and training proportion $p_{\text{train}} = 1.0$. Lower is better.

A.2.1. Neural snapshots at other splits

Tables 5 and 6 report neural FVI snapshot performance under sample splitting for the additional training proportions $p_{\text{train}} = 0.5$ and $p_{\text{train}} = 0.75$. Together, they show how the effect of calibration evolves over training and how it depends on the amount of data reserved for fitting versus calibration.

Iter	Full Raw	Iso	Histogram	Hybrid Iso
10	1.309±0.06	**0.581±0.05**	0.582±0.05	0.581±0.05
25	0.491±0.13	**0.456±0.06**	0.456±0.06	**0.456±0.06**
50	**0.320±0.07**	0.381±0.05	0.382±0.05	0.382±0.05
100	**0.316±0.05**	0.350±0.05	0.351±0.05	0.350±0.05

Table 5. Neural snapshot performance with sample splitting for $n = 50{,}000$ and training proportion $p_{\text{train}} = 0.5$. Lower is better.

Iteration	Full Raw	Iso	Histogram	Hybrid Iso
10	1.28±0.05	0.52±0.05	**0.52±0.05**	0.52±0.05
25	0.44±0.12	**0.38±0.03**	0.38±0.04	0.38±0.04
50	0.34±0.06	**0.33±0.04**	0.33±0.04	0.33±0.04
100	0.33±0.07	**0.31±0.04**	0.31±0.04	0.31±0.04

Table 6. Neural FVI snapshot results for $n = 50{,}000$ with $p_{\text{train}} = 0.75$. Lower is better.

A.2.2. Comparing to the uncalibrated (raw) model without retraining on the full data

Table 7 compares calibrated estimators to the corresponding raw model fit only on the training split, so any improvements reflect calibration rather than retraining on the full dataset. This comparison is relevant when retraining on all available data is costly or when a gold-standard calibration dataset is available, such as recent trajectories or on-policy data, alongside a larger historical dataset used for initial fitting. Table 8 further examines this comparison for the neural method across training iterations, illustrating how the effect of calibration evolves over the course of fitting.

B. Stationary Measures and Contraction Results

Lemma 1 (Bellman contraction under a stationary measure)**.** *Let P be a Markov operator with stationary distribution μ (i.e., $\mu = \mu P$), and define the Bellman operator $\mathcal{T} f := r + \gamma P f$. Then P is nonexpansive in $L^2(\mu)$:*

$$\|P(f-g)\|_{2,\mu} \;\le\; \|f-g\|_{2,\mu}.$$

Consequently, $\mathcal{T}$ is a γ-contraction:

$$\|\mathcal{T}(f-g)\|_{2,\mu} \;\le\; \gamma\,\|f-g\|_{2,\mu}.$$

n	Model	Raw	Iso	Hybrid Iso
10,000	Boost	0.681±0.1	**0.671±0.1**	0.697±0.2
	Lin	0.640±0.04	**0.612±0.05**	0.641±0.05
	Neur	0.582±0.2	0.550±0.2	**0.520±0.1**
50,000	Boost	0.433±0.06	**0.429±0.06**	0.438±0.08
	Lin	0.625±0.02	**0.614±0.03**	0.616±0.03
	Neur	0.419±0.1	0.383±0.08	**0.374±0.08**
100,000	Boost	0.360±0.05	0.358±0.05	**0.353±0.07**
	Lin	0.616±0.02	**0.605±0.02**	0.606±0.02
	Neur	0.379±0.09	0.351±0.05	**0.342±0.05**

Table 7. Main results with sample splitting at training proportion $p_{\text{train}} = 0.5$. The raw estimator is fit only on the training split to isolate the effect of calibration.

Iter	Raw	Iso	Quantile	Hybrid Iso
10	1.424±0.1	1.330±0.1	**0.735±0.2**	0.737±0.2
25	0.665±0.1	0.646±0.1	0.577±0.1	**0.571±0.1**
50	0.612±0.2	0.585±0.2	0.555±0.1	**0.547±0.1**
100	0.582±0.2	0.550±0.2	0.524±0.1	**0.520±0.1**

Table 8. Neural FVI snapshot performance with sample splitting at training proportion $p_{\text{train}} = 0.5$. The raw estimator is fit using only the training split, so differences reflect the effect of calibration rather than access to additional data.

Proof. Because μ is stationary for P, the operator P is nonexpansive in $L^2(\mu)$. Jensen's inequality implies

$$(Ph)^2 \le P(h^2) \qquad \text{pointwise.}$$

Integrating both sides with respect to μ and using $\mu = \mu P$ yields

$$\|Ph\|_{2,\mu}^2 = \int (Ph)^2 \, d\mu \le \int P(h^2) \, d\mu = \int h^2 \, d\mu = \|h\|_{2,\mu}^2.$$

Now apply this with $h = f - g$. Since the reward function r cancels,

$$\mathcal{T} f - \mathcal{T} g = \gamma \, P(f - g).$$

Therefore,

$$\|\mathcal{T} f - \mathcal{T} g\|_{2,\mu} = \gamma \|P(f - g)\|_{2,\mu} \le \gamma \|f - g\|_{2,\mu},$$

which proves the claim. □

Lemma 2 (Stationarity under coarsening)**.** *Let P be a Markov kernel on $\mathcal{S}$ and let ρ be stationary for P (i.e., $\rho P = \rho$). For any measurable coarsening map $g : \mathcal{S} \to \mathbb{R}$, define*

$$P_g f(s) := \mathbb{E}[\, f(S') \mid g(S) = g(s) \,],$$

where (S, S') has joint law $\rho(ds)\, P(s, ds')$. Then ρ is stationary for P_g.

Proof. Let $(S, S') \sim \rho(ds)\, P(s, ds')$, so $S \sim \rho$ and $S' \mid S \sim P(S, \cdot)$. Then

$$\begin{aligned} \int P_g f(s) \, \rho(ds) &= \mathbb{E}_{\rho,P}[P_g f(S)] \\ &= \mathbb{E}_{\rho,P}[\mathbb{E}_{\rho,P}[f(S') \mid g(S)]] \\ &= \mathbb{E}_{\rho,P}[f(S')]. \end{aligned}$$

Since ρ is stationary for P, the marginal of S' is again ρ, hence $\mathbb{E}_{\rho,P}[f(S')] = \int f(s) \, \rho(ds)$. □

C. Proof of Theorem 1

Proof of Theorem 1. Define $P_{\pi,\hat{v}} := \Pi_{\hat{v}} P_\pi$ and $r_{\pi,\hat{v}} := \Pi_{\hat{v}} r_\pi$. Because v_0 and $\hat{v}_0$ satisfy the fixed-point equations

$$v_0 = \mathcal{T}_\pi v_0, \qquad \hat{v}_0 = \mathcal{T}_{\pi,\hat{v}} \hat{v}_0 = r_{\pi,\hat{v}} + \gamma \, P_{\pi,\hat{v}} \hat{v}_0,$$

we begin by writing

$$\hat{v}_0 - v_0 = \big(\mathcal{T}_{\pi,\hat{v}} \hat{v}_0 - \mathcal{T}_{\pi,\hat{v}} v_0\big) + \big(\mathcal{T}_{\pi,\hat{v}} v_0 - \mathcal{T}_\pi v_0\big).$$

Using linearity of the conditional expectation in $\mathcal{T}_{\pi,\hat{v}}$ on differences,

$$\mathcal{T}_{\pi,\hat{v}} \hat{v}_0 - \mathcal{T}_{\pi,\hat{v}} v_0 = \gamma \, P_{\pi,\hat{v}} (\hat{v}_0 - v_0),$$

so

$$\hat{v}_0 - v_0 = \gamma \, P_{\pi,\hat{v}} (\hat{v}_0 - v_0) + \big(\mathcal{T}_{\pi,\hat{v}} v_0 - v_0\big),$$

where we used $\mathcal{T}_\pi v_0 = v_0$ for the second term.

Taking $L^2(\mu_{\hat{v}})$ norms and applying the triangle inequality,

$$\begin{aligned} \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \gamma \, &\|P_{\pi,\hat{v}} (\hat{v}_0 - v_0)\|_{2,\mu_{\hat{v}}} \\ &+ \|\mathcal{T}_{\pi,\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}. \end{aligned}$$

By Lemma 1,

$$\|P_{\pi,\hat{v}} h\|_{2,\mu_{\hat{v}}} \le \|h\|_{2,\mu_{\hat{v}}}.$$

Applying this with $h = \hat{v}_0 - v_0$ gives

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \gamma \, \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} + \|\mathcal{T}_{\pi,\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}.$$

Note that $\mathcal{T}_{\pi,\hat{v}} v = \Pi_{\hat{v}} \mathcal{T}_\pi v$, where $\Pi_{\hat{v}}$ is the L^2 projection onto functions of $\hat{v}$. Thus $\mathcal{T}_{\pi,\hat{v}} v_0 = \Pi_{\hat{v}} v_0$, and therefore

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \gamma \, \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} + \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}.$$

Rearranging,

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \frac{1}{1-\gamma} \, \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}}. \qquad (\star)$$

By the assumed norm comparison,

$$\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\mu_{\hat{v}}} \le \kappa_{\hat{v}} \|\Pi_{\hat{v}} v_0 - v_0\|_{2,\rho}.$$

Combining with $(\star)$ gives

$$\|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}} \le \frac{\kappa_{\hat{v}}}{1-\gamma}\,\|\Pi_{\hat{v}} v_0 - v_0\|_{2,\rho}.$$

Finally, by the triangle inequality,

$$\|\hat{v} - v_0\|_{2,\mu_{\hat{v}}} \le \|\hat{v} - \hat{v}_0\|_{2,\mu_{\hat{v}}} + \|\hat{v}_0 - v_0\|_{2,\mu_{\hat{v}}},$$

which yields the stated result. □

D. Proof of Theorem 2

Proof of Theorem 2. Fix v and treat the nuisance estimates as fixed functions. By definition,

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) = \mathbb{E}\Big[(\pi\hat{q}_v)(S) + \widehat{w}_\pi(A \mid S)\big\{R + \gamma\, v(S') - \hat{q}_v(S,A)\big\} \,\Big|\, S = s\Big].$$

Since $(\pi\hat{q}_v)(S)$ depends only on S,

$$\mathbb{E}\big[(\pi\hat{q}_v)(S) \mid S = s\big] = (\pi\hat{q}_v)(s).$$

For the second term, using the definition of b_0 and $q_v := r_0 + \gamma P v$,

$$\begin{aligned}
&\mathbb{E}\Big[\widehat{w}_\pi(A \mid S)\big\{R + \gamma\, v(S') - \hat{q}_v(S,A)\big\} \,\Big|\, S = s\Big] \\
&\quad = b_0\Big\{\widehat{w}_\pi(\cdot \mid s)\,\mathbb{E}\big[R + \gamma\, v(S') - \hat{q}_v(s,\cdot) \mid S = s, A = \cdot\big]\Big\}(s) \\
&\quad = b_0\big\{\widehat{w}_\pi(q_v - \hat{q}_v)\big\}(s).
\end{aligned}$$

Hence

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) = (\pi\hat{q}_v)(s) + b_0\big\{\widehat{w}_\pi(q_v - \hat{q}_v)\big\}(s).$$

Subtracting $\mathcal{T}_\pi(v)(s) = (\pi q_v)(s)$ yields

$$\begin{aligned}
\widehat{\mathcal{T}}_{0,\pi}(v)(s) - \mathcal{T}_\pi(v)(s) &= (\pi\hat{q}_v - \pi q_v)(s) + b_0\big\{\widehat{w}_\pi(q_v - \hat{q}_v)\big\}(s) \\
&= (\pi(\hat{q}_v - q_v))(s) - b_0\big\{\widehat{w}_\pi(\hat{q}_v - q_v)\big\}(s).
\end{aligned}$$

Using $(\pi f)(s) = b_0\{w_\pi f\}(s)$ with $f = \hat{q}_v - q_v$ gives

$$(\pi(\hat{q}_v - q_v))(s) = b_0\{w_\pi(\hat{q}_v - q_v)\}(s),$$

so

$$\widehat{\mathcal{T}}_{0,\pi}(v)(s) - \mathcal{T}_\pi(v)(s) = b_0\big\{(w_\pi - \widehat{w}_\pi)(\hat{q}_v - q_v)\big\}(s).$$

This proves the claim. □

E. Notation and Maximal Inequalities

Let P_0 denote the joint distribution of (S, A, R, S') induced by the behavior policy, and let P_n denote the empirical measure of the calibration sample $\mathcal{C}_n$.

We define empirical L^2 norms with respect to the state and next-state samples. For any state function f, let

$$\|f\|_{n,S} := \Big(\tfrac{1}{n}\sum_{i=1}^n f(S_i)^2\Big)^{1/2}, \quad \|f\|_{n,S'} := \Big(\tfrac{1}{n}\sum_{i=1}^n f(S_i')^2\Big)^{1/2},$$

where S_i and S_i' denote the observed states and next states, respectively, in the calibration sample $\mathcal{C}_n$.

Further define

$$\widehat{\mathcal{G}} := \Big\{(f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big) : \; f_1, f_2 \in \mathcal{F}_{B,\hat{v}}\Big\}.$$

By assumption, both $\hat{v}$ and $\widehat{\mathcal{T}}_\pi$ are fixed (non-random) operators conditional on the training data, which is independent of the calibration sample $\mathcal{C}_n$. Consequently, the classes $\mathcal{F}_{B,\hat{v}}$ and $\widehat{\mathcal{G}}$ are non-random conditional on the training dataset.

For any distribution Q and any uniformly bounded function class $\mathcal{F}$, let $N(\varepsilon, \mathcal{F}, L^2(Q))$ denote the ε-covering number of $\mathcal{F}$ under the $L^2(Q)$ norm (Van Der Vaart & Wellner, 1996). Define the uniform entropy integral of $\mathcal{F}$ by

$$\mathcal{J}(\delta, \mathcal{F}) := \int_0^\delta \sup_Q \sqrt{\log N(\epsilon, \mathcal{F}, L^2(Q))}\, d\epsilon,$$

where the supremum is taken over all discrete probability distributions Q.

Finally, for two quantities x and y, we write $x \lessapprox y$ to mean that x is bounded above by y up to a universal constant that depends only on global constants appearing in our conditions.

E.1. Local maximal inequality

Let $O_1, \ldots, O_n \in \mathcal{O}$ be independent random variables. For any function $f : \mathcal{O} \to \mathbb{R}$, define

$$\|f\|_{2,hob} := \sqrt{\frac{1}{n}\sum_{i=1}^n \mathbb{E}[f(O_i)^2]}. \tag{7}$$

For a star-shaped class of functions $\mathcal{F}$ and a radius $\delta \in (0, \infty)$, define the localized Rademacher complexity

$$\mathcal{R}_n(\mathcal{F}, \delta) := \mathbb{E}\left[\sup_{\substack{f \in \mathcal{F} \\ \|f\|_{2,hob} \le \delta}} \frac{1}{n}\sum_{i=1}^n \epsilon_i f(O_i)\right],$$

where ϵ_i are i.i.d. Rademacher random variables.

The following lemma provides a local maximal inequality and restates Lemma 11 of (Foster & Syrgkanis, 2023).

Lemma 3 (Local maximal inequality)**.** *Let $\mathcal{F}$ be a star-shaped class of functions satisfying $\sup_{f \in \mathcal{F}} \|f\|_\infty \le M$. Let $\delta > 0$ satisfy the critical radius condition $\mathcal{R}_n(\mathcal{F}, \delta) \le \delta^2$. Suppose further that $n^{-1/2}\sqrt{\log\log(1/\delta)} = o(\delta)$. Then there exists a universal constant $C > 0$ such that, for all $u \ge 1$, with probability at least $1 - e^{-u^2}$, every $f \in \mathcal{F}$ satisfies*

$$\frac{1}{n}\sum_{i=1}^n \big(f(O_i) - \mathbb{E}[f(O_i)]\big) \le C\Big(\delta^2 + \delta\|f\|_{2,hob} + \frac{u\|f\|_{2,hob}}{\sqrt{n}} + \frac{Mu^2}{n}\Big).$$

The following lemma bounds the localized Rademacher complexity in terms of the uniform entropy integral and is a direct consequence of Theorem 2.1 of Van Der Vaart & Wellner (2011).

Lemma 4. *Let $\mathcal{F}$ be a star-shaped class of functions such that $\sup_{f\in\mathcal{F}} \|f\|_\infty \le M$. Then, for every $\delta > 0$,*

$$\mathcal{R}_n(\mathcal{F},\delta) \lesssim \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta,\mathcal{F})\left(1+\frac{\mathcal{J}(\delta,\mathcal{F})}{\delta\sqrt{n}}\right),$$

where the implicit constant depends only on M.

Proof. This bound follows directly from the argument in the proof of Theorem 2.1 of Van Der Vaart & Wellner (2011); see in particular the step where the local Rademacher complexity is controlled by the uniform entropy integral for star-shaped classes. □

F. Proof of Theorem 3

F.1. Technical lemmas

Lemma 5. *Under our conditions,*

$$\mathcal{J}\left(\delta,\widehat{\mathcal{G}}\right) \lesssim \delta\sqrt{B\log(1/\delta)},$$

where the implicit constant is independent of B.

Proof. By assumption, $\|\widehat{P}f-\widehat{P}g\|_{n,A,S} \le L\|f-g\|_{n,S'}$ almost surely. For each s, by Jensen's inequality,

$$|\pi h(s)|^2 = \Big|\sum_a \pi(a\mid s)h(a,s)\Big|^2 \le \sum_a \pi(a\mid s)|h(a,s)|^2,$$

so $\|\pi h\|_{n,S} \le \|h\|_{n,A,S}$. Hence, we also have

$$\|\pi\widehat{P}f-\pi\widehat{P}g\|_{n,S} \le \|\widehat{P}f-\widehat{P}g\|_{n,A,S} \le L\|f-g\|_{n,S'}.$$

Hence, by boundedness of all nuisances,

$$\|\widehat{\mathcal{T}}_\pi(f)-\widehat{\mathcal{T}}_\pi(g)\|_{L^2(P_n)} \lesssim \|f-g\|_{n,S}+\|f-g\|_{n,S'}.$$

Take $f,g\in\widehat{\mathcal{G}}$ with $f=(f_1-f_2)\big(\widehat{\mathcal{T}}_\pi(f_2)-f_2\big)$ and $g=(g_1-g_2)\big(\widehat{\mathcal{T}}_\pi(g_2)-g_2\big)$ for $f_1,f_2,g_1,g_2\in\mathcal{F}_{B,\hat{v}}$. By Lipschitz continuity of multiplication and boundedness of nuisances, we have

$$\|f-g\|_{L^2(P_n)} \lesssim \|f_1-g_1\|_{n,S}+\|f_2-g_2\|_{n,S}+\|f_2-g_2\|_{n,S'}.$$

Taking the supremum over all discrete distributions Q, we obtain

$$\|f-g\|_{L^2(P_n)} \lesssim \sup_Q\|f_1-g_1\|_{L^2(Q)}+\sup_Q\|f_2-g_2\|_{L^2(Q)}.$$

Hence, by preservation of entropy integrals in Van Der Vaart & Wellner (1996),

$$\log N(\varepsilon,\widehat{\mathcal{G}},L^2(P_n)) \lesssim \sup_Q \log N(\varepsilon,\mathcal{F}_{B,\hat{v}},L^2(Q)).$$

Taking the supremum over discrete distributions Q on both sides yields the uniform covering number bound

$$\sup_Q \log N(\varepsilon,\widehat{\mathcal{G}},L^2(Q)) \lesssim \sup_Q \log N(\varepsilon,\mathcal{F}_{B,\hat{v}},L^2(Q)).$$

The class $\mathcal{F}_{B,\hat{v}}$ satisfies

$$\sup_Q \log N\big(\varepsilon,\mathcal{F}_{B,\hat{v}},L^2(Q)\big) \le \sup_Q \log N\big(\varepsilon,\widetilde{\mathcal{F}}_B,L^2(Q\circ\hat{v}^{-1})\big),$$

where $Q\circ\hat{v}^{-1}$ denotes the pushforward of Q under $\hat{v}$. Hence,

$$\sup_Q \log N\big(\varepsilon,\mathcal{F}_{B,\hat{v}},L^2(Q)\big) \le \sup_Q \log N\big(\varepsilon,\widetilde{\mathcal{F}}_B,L^2(Q)\big),$$

where, by a slight abuse of notation, the supremum on the right-hand side is taken over all discrete probability distributions Q on $\mathbb{R}$. The class $\widetilde{\mathcal{F}}_B$ consists of all piecewise-constant functions on $\mathbb{R}$ taking at most B values. Therefore, $\widetilde{\mathcal{F}}_B$ has VC–subgraph dimension $O(B)$, and Van Der Vaart & Wellner (1996) implies

$$\begin{aligned}\sup_Q \log N(\varepsilon,\mathcal{F}_{B,\hat{v}},L^2(Q)) &\lesssim \sup_Q \log N\big(\varepsilon,\widetilde{\mathcal{F}}_B,L^2(Q)\big)\\ &\lesssim B\log(1/\varepsilon).\end{aligned}$$

Consequently, $\mathcal{J}(\delta,\mathcal{F}_{B,\hat{v}}) \lesssim \delta\sqrt{B\log(1/\delta)}$. □

F.2. Proof of Theorem 3

Proof of Theorem 3. Denote $O_i := (S_i,A_i,R_i,S_i')$. The first-order optimality conditions for $\theta_n^{(K)}$ imply that, for each bin $b\in[B]$,

$$\frac{1}{n}\sum_{i=1}^n \mathbf{1}_{I_b}(\hat{v}(S_i))\big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i)-\hat{v}^{(K)}(S_i)\big\}=0.$$

Hence, for any $f\in\mathcal{F}_B$, which is a linear combination of these indicators,

$$\frac{1}{n}\sum_{i=1}^n f(\hat{v}(S_i))\big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i)-\hat{v}^{(K)}(S_i)\big\}=0. \quad (8)$$

Moreover, for any function $g:\mathbb{R}\to\mathbb{R}$, we may write $g\circ\hat{v}^{(K)}=f\circ\hat{v}$ for some f, and therefore, for all such g,

$$\frac{1}{n}\sum_{i=1}^n g(\hat{v}^{(K)}(S_i))\big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i)-\hat{v}^{(K)}(S_i)\big\}=0.$$

Noting that $\Gamma_0(\hat{v}^{(K)})-\hat{v}^{(K)}$ is of the form $g\circ\hat{v}^{(K)}$ for some function g, we have

$$\begin{aligned}0=\frac{1}{n}\sum_{i=1}^n &\Big\{\Gamma_0(\hat{v}^{(K)})(S_i)-\hat{v}^{(K)}(S_i)\Big\}\\ &\times\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i)-\hat{v}^{(K)}(S_i)\Big\}.\end{aligned}$$

Adding and subtracting P_0 yields

$$\begin{aligned} &P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\} \\ &= (P_0 - P_n)\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\Big\} \\ &\quad + P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})\Big\}. \end{aligned} \tag{9}$$

We rewrite the left-hand side of (9). Adding and subtracting and applying the law of total expectation, the calibration error decomposes as

$$\begin{aligned} &\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|^2 \\ &= P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\} \\ &\quad + P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\Gamma_0(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K)})\Big\}. \end{aligned} \tag{10}$$

The first term on the right-hand side can be decomposed as in (9). By the law of total expectation, the proof of Theorem 2, and the Cauchy–Schwarz inequality, the second term satisfies

$$\begin{aligned} &\Big|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \Gamma_0(\hat{v}^{(K)})\Big\}\Big| \\ &= \Big|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \mathcal{T}_\pi(\hat{v}^{(K)})\Big\}\Big| \\ &\le \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \, \big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\big\|_{2,\rho}. \end{aligned}$$

Hence, (9) and (10) together imply that

$$\begin{aligned} &\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|^2 \\ &\le (P_0 - P_n)\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\Big\} \\ &\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \, \big\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\big\| \\ &\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \, \big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\big\|_{2,\rho}. \end{aligned} \tag{11}$$

where $\widehat{\mathcal{T}}_{0,\pi}(v)(s) := \mathbb{E}[\widehat{\mathcal{T}}_\pi(v)(S) \mid S = s]$. Here, the second term on the right-hand side follows from (9), noting that

$$\begin{aligned} &\Big|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})\Big\}\Big| \\ &= \Big|P_0\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\,\mathbb{E}\Big[\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) \mid S\Big]\Big| \\ &\le \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \, \Big\|\mathbb{E}\Big[\widehat{\mathcal{T}}_\pi(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) \mid S\Big]\Big\| \\ &= \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \, \big\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\big\|_{2,\rho}. \end{aligned}$$

We now turn to bounding the empirical process term on the right-hand side of (11). Observe that

$$\big(\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\big)\big(\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\big)$$

lies in a uniformly bounded subset of the class

$$\widehat{\mathcal{G}} := \big\{(f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big) : f_1, f_2 \in \mathcal{F}_{B,\hat{v}}\big\},$$

By Lemma 5, it holds that $\mathcal{J}\left(\delta, \widehat{\mathcal{G}}\right) \lesssim \delta\sqrt{B\log(1/\delta)}$. By assumption, $\widehat{\mathcal{G}}$ is a fixed, nonrandom function class conditional on the training data, which is independent of the calibration sample $\mathcal{C}_n$. Hence, applying Lemma 4 conditional on the training data, we obtain

$$\begin{aligned} \mathcal{R}_n(\widehat{\mathcal{G}}, \delta) &\lesssim \frac{1}{\sqrt{n}}\,\mathcal{J}(\delta, \widehat{\mathcal{G}})\left(1 + \frac{\mathcal{J}(\delta, \widehat{\mathcal{G}})}{\delta\sqrt{n}}\right) \\ &\lesssim \frac{1}{\sqrt{n}}\,\delta\sqrt{B\log(1/\delta)}\left(1 + \frac{\delta\sqrt{B\log(1/\delta)}}{\delta\sqrt{n}}\right) \\ &\lesssim \frac{\delta\sqrt{B\log(1/\delta)}}{\sqrt{n}} + \frac{B\log(1/\delta)}{n}. \end{aligned}$$

The critical radius

$$\delta_n := \sqrt{\frac{B}{n}\,\log\left(\frac{n}{B}\right)},$$

satisfies

$$\delta_n = \inf\{\delta > 0 : \mathcal{R}_n(\widehat{\mathcal{G}}, \delta) \lesssim \delta^2\}.$$

Applying Lemma 3 conditional on the training data with $\mathcal{F} := \widehat{\mathcal{G}}$, we conclude that the following holds with probability at least $1 - e^{-u^2}$ for every $f \in \widehat{\mathcal{G}}$:

$$\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \lesssim \delta_n^2 + \delta_n\|f\|_{2,hob} + \frac{u\|f\|_{2,hob}}{\sqrt{n}} + \frac{u^2}{n}.$$

Choosing $u = \sqrt{\log(1/\eta)}$ gives $1 - e^{-u^2} = 1 - \eta$. Hence, with probability at least $1 - \eta$,

$$\begin{aligned} &\frac{1}{n}\sum_{i=1}^{n}\big(f(O_i) - \mathbb{E}[f(O_i)]\big) \\ &\lesssim \delta_n^2 + \delta_n\|f\|_{2,hob} + \frac{\sqrt{\log(1/\eta)}\,\|f\|_{2,hob}}{\sqrt{n}} + \frac{\log(1/\eta)}{n}. \end{aligned}$$

By boundedness of our nuisances,

$$\big\|\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\}\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\}\big\| \lesssim \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|_{2,\rho}.$$

Hence, with probability at least $1 - \eta$,

$$\begin{aligned} &(P_n - P_0)\Big\{\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\Big\}\Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\Big\} \\ &\quad \lesssim \delta_n^2 + \delta_n\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \\ &\qquad + \frac{\sqrt{\log(1/\eta)}\,\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|}{\sqrt{n}} + \frac{\log(1/\eta)}{n}, \end{aligned}$$

where the implicit constants do not depend on B.

Combining the above with (11), we find with probability at least $1-\eta$,

$$\begin{aligned}
&\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\|^2 \\
&\lesssim \delta_n^2 \;+\; \delta_n \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \\
&\quad + \frac{\sqrt{\log(1/\eta)}}{\sqrt{n}} \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \;+\; \frac{\log(1/\eta)}{n} \\
&\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\| \\
&\quad + \|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho}.
\end{aligned} \tag{12}$$

The inequality in (12) implies that, with probability at least $1-\eta$, the calibration error satisfies

$$\begin{aligned}
\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \;\lesssim\; \delta_n \;+\; \sqrt{\frac{\log(1/\eta)}{n}} \\
+\; \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\| \\
+\; \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho}.
\end{aligned}$$

Recall that $\delta_n := \sqrt{\frac{B}{n}\,\log\left(\frac{n}{B}\right)}$. Then, with probability at least $1-\eta$,

$$\begin{aligned}
\|\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\| \;\lesssim\; \sqrt{\frac{B}{n}\,\log\left(\frac{n}{B}\right)} \;+\; \sqrt{\frac{\log(1/\eta)}{n}} \\
+\; \left\|\widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K)}) - \widehat{\mathcal{T}}_{0,\pi}(\hat{v}^{(K-1)})\right\| \\
+\; \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(K)}} - q_{\hat{v}^{(K)}})\right\|_{2,\rho}.
\end{aligned}$$

□

G. Proof of Theorem 4

G.1. Additional notation

Define the bin–index map $\hat{b}(s)$ by setting $\hat{b}(s) = b$ whenever $\hat{v}(s) \in I_b$. Let $\Pi_{\hat{v},B}$ denote the binning projection

$$(\Pi_{\hat{v},B} g)(s) := E\Big[g(S) \,\Big|\, \hat{v}(S) \in I_{\hat{b}(s)}\Big].$$

Denote

$$\mathcal{T}_{\pi,\hat{v},B}(g)(s) := (\Pi_{\hat{v},B}\mathcal{T}_\pi(g)(s)$$

Define $\hat{v}_{0,B}$ denote the fixed point satisfying

$$\mathcal{T}_{\pi,\hat{v},B}(\hat{v}_{0,B}) = \hat{v}_{0,B}.$$

Such a fixed point exists by Banach's fixed point theorem because $\mathcal{T}_{\pi,\hat{v},B}(g) \in \mathcal{F}_{B,\hat{v}}$ whenever $g \in \mathcal{F}_{B,\hat{v}}$, and the operator is a γ-contraction in the sup-norm.

Recall the projected transition operator

$$(P_{\pi,\hat{v},B}h)(s) = \mathbb{E}_\pi\left[h(S') \,\middle|\, \hat{v}(S) \in I_{\hat{b}(s)}\right].$$

It holds that

$$P_{\pi,\hat{v},B} \;:=\; \Pi_{\hat{v},B} P_\pi,$$

and

$$\Pi_{\hat{v},B}\mathcal{T}_\pi(f) - \Pi_{\hat{v},B}\mathcal{T}_\pi(g) = \gamma\, P_{\pi,\hat{v},B}(f-g).$$

G.2. Supporting lemmas

We begin by bounding the error between the projected fixed point $\hat{v}_{0,B}$ and the true value function v_0.

Lemma 6 (Approximation error for the projected fixed point)**.** *Assume that $\mu_{\hat{v},B}$ is stationary for $P_{\pi,\hat{v},B}$. Then*

$$\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} \;\le\; \frac{1}{1-\gamma}\,\|(I - \Pi_{\hat{v},B})v_0\|_{L^2(\mu_{\hat{v},B})}.$$

Proof. Because v_0 and $\hat{v}_{0,B}$ satisfy $v_0 = \mathcal{T}_\pi v_0$ and $\hat{v}_{0,B} = \Pi_{\hat{v},B}\mathcal{T}_\pi\hat{v}_{0,B}$, we have

$$\hat{v}_{0,B} - v_0 = \Pi_{\hat{v},B}\big(\mathcal{T}_\pi\hat{v}_{0,B} - \mathcal{T}_\pi v_0\big) + (I - \Pi_{\hat{v},B})\mathcal{T}_\pi v_0.$$

Since $\mathcal{T}_\pi(f) - \mathcal{T}_\pi(g) = \gamma P_\pi(f-g)$ and $\Pi_{\hat{v},B}$ is affine but linear on differences, the first term equals

$$\begin{aligned}
\Pi_{\hat{v},B}\big(\mathcal{T}_\pi\hat{v}_{0,B} - \mathcal{T}_\pi v_0\big) &= \gamma\,\Pi_{\hat{v},B}P_\pi(\hat{v}_{0,B} - v_0) \\
&= \gamma\, P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0).
\end{aligned}$$

Using $\mathcal{T}_\pi v_0 = v_0$, we therefore obtain the error identity

$$\hat{v}_{0,B} - v_0 = \gamma\, P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0) + (I - \Pi_{\hat{v},B})v_0.$$

Taking $L^2(\mu_{\hat{v},B})$ norms and applying the triangle inequality yields

$$\begin{aligned}
\|\hat{v}_{0,B} - v_0\|_{2,\mu_{\hat{v},B}} \le \gamma\,\|P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0)\|_{2,\mu_{\hat{v},B}} \\
+\; \|(I - \Pi_{\hat{v},B})v_0\|_{2,\mu_{\hat{v},B}}.
\end{aligned}$$

Because $P_{\pi,\hat{v},B}$ is a Markov operator and $\mu_{\hat{v},B}$ is stationary for $P_{\pi,\hat{v},B}$, it is nonexpansive in $L^2(\mu_{\hat{v},B})$. By Jensen's inequality, $(P_{\pi,\hat{v},B}h)^2 \le P_{\pi,\hat{v},B}(h^2)$ pointwise, hence

$$\begin{aligned}
\|P_{\pi,\hat{v},B}h\|^2_{L^2(\mu_{\hat{v},B})} &= \int (P_{\pi,\hat{v},B}h)^2\, d\mu_{\hat{v},B} \\
&\le \int P_{\pi,\hat{v},B}(h^2)\, d\mu_{\hat{v},B} \\
&= \int h^2\, d\mu_{\hat{v},B} \\
&= \|h\|^2_{L^2(\mu_{\hat{v},B})}.
\end{aligned}$$

where the equality uses stationarity of $\mu_{\hat{v},B}$. Applying this with $h = \hat{v}_{0,B} - v_0$ gives $\|P_{\pi,\hat{v},B}(\hat{v}_{0,B} - v_0)\|_{L^2(\mu_{\hat{v},B})} \leq \|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})}$. Therefore,

$$\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} \leq \gamma \|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} + \|(I - \Pi_{\hat{v},B})v_0\|_{L^2(\mu_{\hat{v},B})}$$

Rearranging completes the proof:

$$\|\hat{v}_{0,B} - v_0\|_{L^2(\mu_{\hat{v},B})} \leq \frac{1}{1-\gamma} \|(I - \Pi_{\hat{v},B})v_0\|_{L^2(\mu_{\hat{v},B})}.$$

□

Lemma 7 (Inexact iterations of $\mathcal{T}_{\pi,\hat{v},B}$)**.** *Assume $\mu_{\hat{v},B}$ is stationary for $P_{\pi,\hat{v},B}$, so $\mu_{\hat{v},B} = \mu_{\hat{v},B} P_{\pi,\hat{v},B}$. Let $\{\eta_k\}_{k\geq 1}$ be any sequence such that*

$$\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)})\|_{2,\mu_{\hat{v},B}} \leq \eta_k \qquad \textit{for all } k.$$

Then, for any $K \geq 1$,

$$\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \leq \gamma^K \|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K} \gamma^{K-j} \eta_j.$$

Proof. Because $\mathcal{F}_B$ consists of functions that are constant on each bin, conditioning on $\hat{b}(S)$ yields a function that is also constant on bins. Thus $\mathcal{T}_{\pi,\hat{v},B}$ maps $\mathcal{F}_B$ into itself.

Define the per-iteration error function

$$e_k := \hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)}), \qquad \|e_k\|_{2,\mu_{\hat{v},B}} \leq \eta_k.$$

Since $\hat{v}^{(k)}, \hat{v}_{0,B} \in \mathcal{F}_B$, there exist $\hat{\theta}^{(k)}$ and $\theta^\star$ such that $\hat{v}^{(k)} = \hat{\theta}^{(k)} \circ \hat{v}$ and $\hat{v}_{0,B} = \theta^\star \circ \hat{v}$. By Lemma 1, $\mathcal{T}_{\pi,\hat{v},B}$ is a γ-contraction under $\|\cdot\|_{2,\mu_{\hat{v},B}}$. Applying the standard inexact-iteration bound (e.g., Lemma 4 of (van der Laan et al., 2025b)) gives

$$\|(\hat{\theta}^{(K)} - \theta^\star) \circ \hat{v}\|_{2,\mu_{\hat{v},B}} \leq \gamma^K \|(\hat{\theta}^{(0)} - \theta^\star) \circ \hat{v}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K} \gamma^{K-j} \|e_j\|_{2,\mu_{\hat{v},B}}.$$

Substituting $\hat{v}^{(k)} = \hat{\theta}^{(k)} \circ \hat{v}$ and $\hat{v}_{0,B} = \theta^\star \circ \hat{v}$ yields

$$\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \leq \gamma^K \|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K} \gamma^{K-j} \eta_j.$$

□

Lemma 8 (Error bound for inexact updates)**.** *There exists a $C < \infty$, such that, with probability at least $1 - \delta$, it holds that, uniformly over $k \in [K]$,*

$$\begin{aligned} &\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(v^{(k-1)})\| \\ &\quad \leq \sqrt{\frac{B}{n} \log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}} \\ &\qquad + C\left\|(\widehat{w}_\pi - w_\pi)\left(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}}\right)\right\|_{2,\rho}. \end{aligned}$$

Proof. Equation 8 in the proof of Theorem 3 shows that, for any transformation $f \in \mathcal{F}_B$ and any k,

$$\frac{1}{n}\sum_{i=1}^{n} f(\hat{v}(S_i))\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(k-1)})(O_i) - \hat{v}^{(k)}(S_i)\} = 0.$$

Applying the above with $f \circ \hat{v} = \hat{v}_B^{\star(k)} - \hat{v}^{(k)}$, where

$$\hat{v}_B^{\star(k)} := \mathcal{T}_{\pi,\hat{v},B}\left(\theta^{(k-1)}\right),$$

we obtain

$$\frac{1}{n}\sum_{i=1}^{n} (\hat{v}_B^{\star(k)} - \hat{v}^{(K)})(S_i)\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(K-1)}\right)(O_i) - \hat{v}^{(K)}(S_i)\right\} = 0.$$

Adding and subtracting P_0, we find

$$\begin{aligned} &P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] \\ &= (P_0 - P_n)\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right]. \end{aligned} \tag{13}$$

First, we study the left–hand side of (13). We have

$$\begin{aligned} &P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] \\ &= \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \\ &\quad + P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}_B^{\star(k)}\right\}\right]. \end{aligned}$$

By the law of total expectation,

$$\begin{aligned} &P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}_B^{\star(k)}\right\}\right] \\ &= P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \mathcal{T}_{\pi,\hat{v},B}\left(\theta^{(k-1)}\right)\right\}\right] \\ &= P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \mathcal{T}_\pi\left(\hat{v}^{(k-1)}\right)\right\}\right]. \end{aligned}$$

By Theorem 2 and arguing as in the proof of Theorem 3, it follows that

$$\begin{aligned} &\left|P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}_B^{\star(k)}\right\}\right]\right| \\ &\leq \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| \left\|(\widehat{w}_\pi - w_\pi)\left(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}}\right)\right\|_{2,\rho}. \end{aligned}$$

Putting it all together,

$$\begin{aligned} &\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \leq P_0\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] \\ &+ \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| \left\|(\widehat{w}_\pi - w_\pi)\left(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}}\right)\right\|_{2,\rho}. \end{aligned}$$

Hence, by (13),

$$\begin{aligned} &\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \\ &\quad \leq (P_0 - P_n)\left[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\left\{\widehat{\mathcal{T}}_\pi\left(\hat{v}^{(k-1)}\right) - \hat{v}^{(k)}\right\}\right] \\ &\quad + \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| \left\|(\widehat{w}_\pi - w_\pi)\left(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}}\right)\right\|_{2,\rho}. \end{aligned} \tag{14}$$

Next, we obtain a high–probability bound for the first term right–hand side of (14). Applying Lemmas 3, 4, and 5, and arguing as in the proof of Theorem 3, we obtain that, with probability at least $1-\delta$,

$$\Big|(P_0 - P_n)\Big[(\hat{v}_B^{\star(k)} - \hat{v}^{(k)})\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(k-1)}) - \hat{v}^{(k)}\}\Big]\Big| \lesssim \delta_n^2 + \delta_n \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| + \frac{\sqrt{\log(1/\delta)}\,\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|}{\sqrt{n}} + \frac{\log(1/\delta)}{n},$$

where $\delta_n := \sqrt{(B/n)\,\log(B/\delta)}$, and the implicit constants do not depend on B.

Plugging this high probability bound into (14), we obtain that, with probability at least $1-\delta$,

$$\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|^2 \lesssim \delta_n^2 + \delta_n \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| + \frac{\sqrt{\log(1/\delta)}\,\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|}{\sqrt{n}} + \frac{\log(1/\delta)}{n} + \|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\|\,\big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\big\|_{2,\rho}.$$

Recalling that $\delta_n := \sqrt{\frac{B}{n}\,\log\left(\frac{n}{B}\right)}$, the inequality above implies that, with probability at least $1-\delta$,

$$\|\hat{v}_B^{\star(k)} - \hat{v}^{(k)}\| \lesssim \sqrt{\frac{B}{n}\,\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(1/\delta)}{n}} + \big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\big\|_{2,\rho}.$$

Recalling that $\eta_k = \hat{v}_B^{\star(k)}$, the result follows by a union bound over $k \in [K]$. □

G.3. Proof of Theorem 4

Proof of Theorem 4. For $C \in (0,\infty)$ large enough, define

$$\eta_k = C\left(\sqrt{\frac{B}{n}\,\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) + \big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(k-1)}} - q_{\hat{v}^{(k-1)}})\big\|_{2,\rho}.$$

By Lemma 8 and the norm bound in C4, for C large enough, with probability at least $1-\delta$ and uniformly over $k \in [K]$,

$$\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)})\|_{2,\mu_{\hat{v},B}} \le \hat{\kappa}\,\|\hat{v}^{(k)} - \mathcal{T}_{\pi,\hat{v},B}(\hat{v}^{(k-1)})\| \le \hat{\kappa}\,\eta_k.$$

Define $\tilde{\eta}_k := \hat{\kappa}\,\eta_k$. Then the above shows that the assumption of Lemma 7 holds with $\tilde{\eta}_k$ in place of η_k.

Applying the deterministic inequality of Lemma 7 with $\tilde{\eta}_k$ yields, with probability at least $1-\delta$,

$$\begin{aligned}
&\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\le \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} + \sum_{j=1}^{K}\gamma^{K-j}\,\tilde{\eta}_j \\
&\le \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C\hat{\kappa}}{1-\gamma}\left(\sqrt{\frac{B}{n}\,\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\hat{\kappa}}{1-\gamma}\max_{j\in[K]}\big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\big\|_{2,\rho}.
\end{aligned}$$

By Lemma 6,

$$\|\hat{v}_{0,B} - v_0\|_{2,\mu_{\hat{v},B}} \le \frac{1}{1-\gamma}\,\|(I - \Pi_{\hat{v},B})v_0\|_{2,\mu_{\hat{v},B}}.$$

Combining this with our high–probability bound on $\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}}$, we find that, with probability at least $1-\delta$,

$$\begin{aligned}
&\|\hat{v}^{(K)} - v_0\|_{2,\mu_{\hat{v},B}} \\
&\le \frac{1}{1-\gamma}\|\Pi_{\hat{v},B}v_0 - v_0\|_{2,\mu_{\hat{v},B}} \\
&\quad + \gamma^K\,\|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C\hat{\kappa}}{1-\gamma}\left(\sqrt{\frac{B}{n}\,\log\Big(\frac{n}{B}\Big)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\hat{\kappa}}{1-\gamma}\max_{j\in[K]}\big\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\big\|_{2,\rho}.
\end{aligned}$$

Since $\Pi_{\hat{v},B}$ is a projection under $\|\cdot\|_{2,\rho}$, we have

$$\begin{aligned}
\|\hat{v}_{0,B} - v_0\|_{2,\mu_{\hat{v},B}} &\le \frac{\hat{\kappa}}{1-\gamma}\,\|(I - \Pi_{\hat{v},B})v_0\| \\
&\le \frac{\hat{\kappa}}{1-\gamma}\min_{\theta\in\mathcal{F}_B}\|\theta\circ\hat{v} - v_0\|,
\end{aligned}$$

where we used the second norm bound in C4. □

G.4. Bound on Successive Iteration Errors

Lemma 9 (Bound on Successive Iteration Errors). *Under the conditions of Theorem 4,*

$$\begin{aligned}
&\|\hat{v}^{(K)} - \hat{v}^{(K-1)}\|_{2,\mu_{\hat{v},B}} \\
&\lesssim \gamma^K \|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\kappa_{\hat{v},B}}{1-\gamma} \max_{j\in[K]} \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\right\|_{2,\rho}.
\end{aligned}$$

Proof. The proof of Theorem 4 established that

$$\begin{aligned}
&\|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\leq \gamma^K \|\hat{v} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&\quad + \frac{C\kappa_{\hat{v},B}}{1-\gamma}\left(\sqrt{\frac{B}{n}\log\left(\frac{n}{B}\right)} + \sqrt{\frac{\log(K/\delta)}{n}}\right) \\
&\quad + \frac{\kappa_{\hat{v},B}}{1-\gamma} \max_{j\in[K]} \left\|(\widehat{w}_\pi - w_\pi)\,(\widehat{q}_{\hat{v}^{(j-1)}} - q_{\hat{v}^{(j-1)}})\right\|_{2,\rho}.
\end{aligned}$$

By the triangle inequality,

$$\begin{aligned}
\|\hat{v}^{(K)} - \hat{v}^{(K-1)}\|_{2,\mu_{\hat{v},B}} \leq\ & \|\hat{v}^{(K-1)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}} \\
&+ \|\hat{v}^{(K)} - \hat{v}_{0,B}\|_{2,\mu_{\hat{v},B}}.
\end{aligned}$$

The result follows by applying the above bound with $K-1$ and K. □

H. Proofs of Theorems 5 and 6

Proof of Theorem 5. The first-order optimality conditions for isotonic regression (equivalently, its interpretation as a histogram estimator) imply that for any function $f : \mathbb{R} \to \mathbb{R}$ that is a linear combination of the indicator functions defining the isotonic partition,

$$\frac{1}{n}\sum_{i=1}^{n} f\big(\hat{v}^{(K)}(S_i)\big)\Big\{\widehat{\mathcal{T}}_\pi\big(\hat{v}^{(K-1)}\big)(O_i) - \hat{v}^{(K)}(S_i)\Big\} = 0.$$

See, for example, the proof of Lemma C.1 in Van Der Laan et al. (2023) and van der Laan et al. (2024a). Choosing f appropriately yields

$$\begin{aligned}
0 = \frac{1}{n}\sum_{i=1}^{n} &\Big\{\Gamma_0(\hat{v}^{(K)})(S_i) - \hat{v}^{(K)}(S_i)\Big\} \\
&\times \Big\{\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)})(O_i) - \hat{v}^{(K)}(S_i)\Big\},
\end{aligned}$$

which is the same basic equality as (9) in the proof of Theorem 3. The remainder of the argument proceeds along the same lines with minor modifications.

Specifically, let $\mathcal{F}_{TV}$ denote the union of $\mathcal{F}_{iso}$, which is uniformly bounded by $2M$ under Condition C1, with all functions of bounded total variation bounded by the constant C in Condition C5. By Van Der Vaart & Wellner (1996), this class satisfies the uniform entropy integral bound $\mathcal{J}(\delta, \mathcal{F}_{TV}) \lesssim \sqrt{\delta}$.

Under Condition C5 and by Lemma 6 of van der Laan et al. (2024a), $\Gamma_0(\hat{v}^{(K)})$ has finite total variation bounded and lies in $\mathcal{F}_{TV,\hat{v}} := \{f \circ \hat{v} : f \in \mathcal{F}_{TV}\}$. Thus, it holds that

$$\big(\Gamma_0(\hat{v}^{(K)}) - \hat{v}^{(K)}\big)\big(\widehat{\mathcal{T}}_\pi(\hat{v}^{(K-1)}) - \hat{v}^{(K)}\big)$$

lies in a uniformly bounded subset of the class

$$\widehat{\mathcal{G}} := \Big\{(f_1 - f_2)\big(\widehat{\mathcal{T}}_\pi(f_2) - f_2\big) :\ f_1, f_2 \in \mathcal{F}_{TV,\hat{v}}\Big\}.$$

Arguing as in the proof of Theorem 3, we have

$$\mathcal{J}(\delta, \widehat{\mathcal{G}}) \lesssim \mathcal{J}(\delta, \mathcal{F}_{TV}) \lesssim \sqrt{\delta}.$$

The proof now follows directly from Theorem 3 with this new choice of $\widehat{\mathcal{G}}$ and its associated critical radius $\delta_n = n^{-1/3}$ for monotone functions. □

Proof of Theorem 6. The result follows directly from a union bound and the proofs of Theorems 3 and 4, with $\mathcal{F}_B$ replaced by $\mathcal{F}_{B_n}$, the class of all piecewise-constant functions with at most B_n constant segments. In particular, the entropy bound in Lemma 5 continues to apply to this class. Notably, the proofs of these theorems allow for data-adaptive partitions and require only a deterministic upper bound on the number of constant segments. □